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Automated exchange of healthcare information for fulfillment of medication doses

Abstract

Automated healthcare information exchange. The healthcare information may comprise one or more dose orders corresponding to dose medications to be administered to a patient. The healthcare information may be received (e.g., from an EMR system such as a hospital information system (HIS) or the like) in the form of a healthcare information data stream. The information may then be standardized in to a standardized intermediate form. For instance, data may be parsed from the data stream and used to populate a staging table. In turn, data from the staging table may be transformed into an input format specific to a given dose fulfillment client to which the dose order is provided for fulfillment of the dose. Additionally, exception processing. logistical processing, and management functionality may be applied to the dose orders.

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5395321	12/1994	Kawahara et al.	N/A	N/A
5398336	12/1994	Tantry et al.	N/A	N/A
5401059	12/1994	Ferrario	N/A	N/A
5404292	12/1994	Hendrickson	N/A	N/A
5404384	12/1994	Colburn et al.	N/A	N/A
5406473	12/1994	Yoshikura et al.	N/A	N/A
5412715	12/1994	Volpe	N/A	N/A
5415167	12/1994	Wilk	N/A	N/A
5416695	12/1994	Stutman et al.	N/A	N/A
5417222	12/1994	Dempsey et al.	N/A	N/A
5420977	12/1994	Sztipanovits et al.	N/A	N/A
5421343	12/1994	Feng	N/A	N/A
5423746	12/1994	Burkett et al.	N/A	N/A

5429602	12/1994	Hauser	N/A	N/A
5431201	12/1994	Torchia et al.	N/A	N/A
5431299	12/1994	Brewer et al.	N/A	N/A
5431627	12/1994	Pastrone et al.	N/A	N/A
5433736	12/1994	Nilsson	N/A	N/A
5438607	12/1994	Przygoda, Jr. et al.	N/A	N/A
5440699	12/1994	Farrand et al.	N/A	N/A
5441047	12/1994	David et al.	N/A	N/A
5445294	12/1994	Gardner et al.	N/A	N/A
5445621	12/1994	Poli et al.	N/A	N/A
5446868	12/1994	Gardea, II et al.	N/A	N/A
5453098	12/1994	Botts et al.	N/A	N/A
5455851	12/1994	Chaco et al.	N/A	N/A
5458123	12/1994	Unger	N/A	N/A
5460294	12/1994	Williams	N/A	N/A
5460605	12/1994	Tuttle et al.	N/A	N/A
5461665	12/1994	Shur et al.	N/A	N/A
5462051	12/1994	Oka et al.	N/A	N/A
5464392	12/1994	Epstein et al.	N/A	N/A
5465286	12/1994	Clare et al.	N/A	N/A
5467773	12/1994	Bergelson et al.	N/A	N/A
5468110	12/1994	McDonald et al.	N/A	N/A
5469855	12/1994	Pompei et al.	N/A	N/A
5471382	12/1994	Tallman et al.	N/A	N/A
5474552	12/1994	Palti	N/A	N/A
5482043	12/1995	Zulauf	N/A	N/A
5482446	12/1995	Williamson et al.	N/A	N/A
5485408	12/1995	Blomquist	N/A	N/A
5490610	12/1995	Pearson	N/A	N/A
5494592	12/1995	Latham, Jr. et al.	N/A	N/A
5496265	12/1995	Langley et al.	N/A	N/A
5496273	12/1995	Pastrone et al.	N/A	N/A
5502944	12/1995	Kraft et al.	N/A	N/A
5507412	12/1995	Ebert et al.	N/A	N/A
5508912	12/1995	Schneiderman	N/A	N/A
5509318	12/1995	Gomes	N/A	N/A
5509422	12/1995	Fukami	N/A	N/A
5513957	12/1995	O'Leary	N/A	N/A
5514088	12/1995	Zakko	N/A	N/A
5514095	12/1995	Brightbill et al.	N/A	N/A
5515426	12/1995	Yacenda et al.	N/A	N/A
5520450	12/1995	Colson, Jr. et al.	N/A	N/A
5520637	12/1995	Pager et al.	N/A	N/A
5522396	12/1995	Langer et al.	N/A	N/A
5522798	12/1995	Johnson et al.	N/A	N/A
5526428	12/1995	Arnold	N/A	N/A
5528503	12/1995	Moore et al.	N/A	N/A
5529063	12/1995	Hill	N/A	N/A
5531680	12/1995	Dumas et al.	N/A	N/A
5531697	12/1995	Olsen et al.	N/A	N/A
5531698	12/1995	Olsen	N/A	N/A
5533079	12/1995	Colburn et al.	N/A	N/A
5533981	12/1995	Mandro et al.	N/A	N/A
5534691	12/1995	Holdaway et al.	N/A	N/A
5536084	12/1995	Curtis et al.	N/A	N/A
5537313	12/1995	Pirelli	N/A	N/A
5537853	12/1995	Finburgh et al.	N/A	N/A
5542420	12/1995	Goldman et al.	N/A	N/A
5544649	12/1995	David et al.	N/A	N/A

5544651	12/1995	Wilk	N/A	N/A
5544661	12/1995	Davis et al.	N/A	N/A
5545140	12/1995	Conero et al.	N/A	N/A
5546580	12/1995	Seliger et al.	N/A	N/A
5547470	12/1995	Johnson et al.	N/A	N/A
5549117	12/1995	Tacklind et al.	N/A	N/A
5549460	12/1995	O'Leary	N/A	N/A
5553609	12/1995	Chen et al.	N/A	N/A
5558638	12/1995	Evers et al.	N/A	N/A
5558640	12/1995	Pfeiler et al.	N/A	N/A
5560352	12/1995	Heim et al.	N/A	N/A
5562232	12/1995	Pearson	N/A	N/A
5562621	12/1995	Claude et al.	N/A	N/A
5563347	12/1995	Martin et al.	N/A	N/A
5564434	12/1995	Halperin et al.	N/A	N/A
5564803	12/1995	McDonald et al.	N/A	N/A
5568912	12/1995	Minami et al.	N/A	N/A
5569186	12/1995	Lord et al.	N/A	N/A
5569187	12/1995	Kaiser	N/A	N/A
5571258	12/1995	Pearson	N/A	N/A
5573502	12/1995	LeCocq et al.	N/A	N/A
5573506	12/1995	Vasko	N/A	N/A
5575632	12/1995	Morris et al.	N/A	N/A
5576952	12/1995	Stutman et al.	N/A	N/A
5579001	12/1995	Dempsey et al.	N/A	N/A
5579378	12/1995	Arlinghaus, Jr.	N/A	N/A
5581369	12/1995	Righter et al.	N/A	N/A
5581687	12/1995	Lyle et al.	N/A	N/A
5582593	12/1995	Hultman	N/A	N/A
5583758	12/1995	Mcilroy et al.	N/A	N/A
5588815	12/1995	Zaleski, II	N/A	N/A
5589932	12/1995	Garcia-Rubio et al.	N/A	N/A
5590648	12/1996	Mitchell et al.	N/A	N/A
5591344	12/1996	Kenley et al.	N/A	N/A
5593267	12/1996	McDonald et al.	N/A	N/A
5594637	12/1996	Eisenberg et al.	N/A	N/A
5594786	12/1996	Chaco et al.	N/A	N/A
5597995	12/1996	Williams et al.	N/A	N/A
5598536	12/1996	Slaughter, III et al.	N/A	N/A
5601445	12/1996	Schipper et al.	N/A	N/A
5609575	12/1996	Larson et al.	N/A	N/A
5609576	12/1996	Voss et al.	N/A	N/A
5613115	12/1996	Gihl et al.	N/A	N/A
5619428	12/1996	Lee et al.	N/A	N/A
5619991	12/1996	Sloane	N/A	N/A
5623652	12/1996	Vora et al.	N/A	N/A
5623925	12/1996	Swenson et al.	N/A	N/A
5626144	12/1996	Tacklind et al.	N/A	N/A
5628619	12/1996	Wilson	N/A	N/A
5630710	12/1996	Tune et al.	N/A	N/A
5631844	12/1996	Margrey et al.	N/A	N/A
5633910	12/1996	Cohen	N/A	N/A
D380260	12/1996	Hyman	N/A	N/A
5634893	12/1996	Rishton	N/A	N/A
5637082	12/1996	Pages et al.	N/A	N/A
5637093	12/1996	Hyman et al.	N/A	N/A
5640301	12/1996	Roecher et al.	N/A	N/A
5640953	12/1996	Bishop et al.	N/A	N/A
5641628	12/1996	Bianchi	N/A	N/A

5643193	12/1996	Papillon et al.	N/A	N/A
5643212	12/1996	Coutre et al.	N/A	N/A
5647853	12/1996	Feldmann et al.	N/A	N/A
5647854	12/1996	Olsen et al.	N/A	N/A
5651775	12/1996	Walker et al.	N/A	N/A
5652566	12/1996	Lambert	N/A	N/A
5658240	12/1996	Urdahl et al.	N/A	N/A
5658250	12/1996	Blomquist et al.	N/A	N/A
5661978	12/1996	Holmes et al.	N/A	N/A
5664270	12/1996	Bell et al.	N/A	N/A
5666404	12/1996	Ciccotelli et al.	N/A	N/A
D385646	12/1996	Chan	N/A	N/A
5678562	12/1996	Sellers	N/A	N/A
5678568	12/1996	Uchikubo et al.	N/A	N/A
5681285	12/1996	Ford et al.	N/A	N/A
5682526	12/1996	Smokoff et al.	N/A	N/A
5683367	12/1996	Jordan et al.	N/A	N/A
5685844	12/1996	Marttila	N/A	N/A
5687717	12/1996	Halpern	N/A	N/A
5687734	12/1996	Dempsey et al.	N/A	N/A
5695473	12/1996	Olsen	N/A	N/A
5697951	12/1996	Harpstead	N/A	N/A
5700998	12/1996	Palti	N/A	N/A
5701894	12/1996	Cherry et al.	N/A	N/A
5704351	12/1997	Mortara et al.	N/A	N/A
5704364	12/1997	Saltzstein et al.	N/A	N/A
5704366	12/1997	Tacklind et al.	N/A	N/A
5712798	12/1997	Langley et al.	N/A	N/A
5712912	12/1997	Tomko et al.	N/A	N/A
5713485	12/1997	Liff et al.	N/A	N/A
5713856	12/1997	Eggers et al.	N/A	N/A
5715823	12/1997	Wood et al.	N/A	N/A
5716114	12/1997	Holmes et al.	N/A	N/A
5716194	12/1997	Butterfield et al.	N/A	N/A
5718562	12/1997	Lawless et al.	N/A	N/A
5719761	12/1997	Gatti et al.	N/A	N/A
RE35743	12/1997	Pearson	N/A	N/A
5724025	12/1997	Tavori	N/A	N/A
5724580	12/1997	Levin et al.	N/A	N/A
5732709	12/1997	Tacklind et al.	N/A	N/A
5733259	12/1997	Valcke et al.	N/A	N/A
5735887	12/1997	Barreras, Sr. et al.	N/A	N/A
5737539	12/1997	Edelson et al.	N/A	N/A
5740185	12/1997	Bosse	N/A	N/A
5740800	12/1997	Hendrickson et al.	N/A	N/A
5745366	12/1997	Higham et al.	N/A	N/A
5745378	12/1997	Barker et al.	N/A	N/A
5752917	12/1997	Fuchs	N/A	N/A
5752976	12/1997	Duffin et al.	N/A	N/A
5755563	12/1997	Clegg et al.	N/A	N/A
5758095	12/1997	Albaum et al.	N/A	N/A
5764923	12/1997	Tallman et al.	N/A	N/A
5766155	12/1997	Hyman et al.	N/A	N/A
5769811	12/1997	Stacey et al.	N/A	N/A
5771657	12/1997	Lasher et al.	N/A	N/A
5772585	12/1997	Lavin et al.	N/A	N/A
5772586	12/1997	Heinonen et al.	N/A	N/A
5772635	12/1997	Dastur et al.	N/A	N/A
5772637	12/1997	Heinzmann et al.	N/A	N/A

5776057	12/1997	Swenson et al.	N/A	N/A
5778345	12/1997	McCartney	N/A	N/A
5778882	12/1997	Raymond et al.	N/A	N/A
5781442	12/1997	Engleson et al.	N/A	N/A
5782805	12/1997	Meinzer et al.	N/A	N/A
5782878	12/1997	Morgan et al.	N/A	N/A
5785650	12/1997	Akasaka et al.	N/A	N/A
5788669	12/1997	Peterson	N/A	N/A
5788851	12/1997	Kenley et al.	N/A	N/A
5790409	12/1997	Fedor et al.	N/A	N/A
5791342	12/1997	Woodard	N/A	N/A
5791880	12/1997	Wilson	N/A	N/A
5793861	12/1997	Haigh	N/A	N/A
5793969	12/1997	Kamentsky et al.	N/A	N/A
5795317	12/1997	Brierton et al.	N/A	N/A
5795327	12/1997	Wilson et al.	N/A	N/A
5797515	12/1997	Liff et al.	N/A	N/A
5800387	12/1997	Duffy et al.	N/A	N/A
5801755	12/1997	Echerer	N/A	N/A
5803906	12/1997	Pratt et al.	N/A	N/A
5805442	12/1997	Crater et al.	N/A	N/A
5805454	12/1997	Valerino et al.	N/A	N/A
5805456	12/1997	Higham et al.	N/A	N/A
5805505	12/1997	Zheng et al.	N/A	N/A
5807321	12/1997	Stoker et al.	N/A	N/A
5807322	12/1997	Lindsey et al.	N/A	N/A
5807336	12/1997	Russo et al.	N/A	N/A
5810747	12/1997	Brudny et al.	N/A	N/A
5812410	12/1997	Lion et al.	N/A	N/A
5814015	12/1997	Gargano et al.	N/A	N/A
5815566	12/1997	Ramot et al.	N/A	N/A
5818528	12/1997	Roth et al.	N/A	N/A
5822418	12/1997	Yacenda et al.	N/A	N/A
5822544	12/1997	Chaco et al.	N/A	N/A
5823949	12/1997	Goltra	N/A	N/A
5826237	12/1997	Macrae et al.	N/A	N/A
5829438	12/1997	Gibbs et al.	N/A	N/A
5832447	12/1997	Rieker et al.	N/A	N/A
5832448	12/1997	Brown	N/A	N/A
5832450	12/1997	Myers et al.	N/A	N/A
5833599	12/1997	Schrier et al.	N/A	N/A
5835897	12/1997	Dang	N/A	N/A
5836910	12/1997	Duffy et al.	N/A	N/A
5841975	12/1997	Layne et al.	N/A	N/A
5842841	12/1997	Danby et al.	N/A	N/A
5842976	12/1997	Williamson	N/A	N/A
5845253	12/1997	Rensimer et al.	N/A	N/A
5848593	12/1997	McGrady et al.	N/A	N/A
5851186	12/1997	Wood et al.	N/A	N/A
5852590	12/1997	De La Huerga	N/A	N/A
5853387	12/1997	Clegg et al.	N/A	N/A
5855550	12/1998	Lai et al.	N/A	N/A
5857967	12/1998	Frid et al.	N/A	N/A
5859972	12/1998	Subramaniam et al.	N/A	N/A
5865745	12/1998	Schmitt et al.	N/A	N/A
5865786	12/1998	Sibalis et al.	N/A	N/A
5867821	12/1998	Ballantyne et al.	N/A	N/A
5871465	12/1998	Vasko	N/A	N/A
5876926	12/1998	Beecham	N/A	N/A

5880443	12/1998	McDonald et al.	N/A	N/A
5882338	12/1998	Gray	N/A	N/A
5883370	12/1998	Walker et al.	N/A	N/A
5883576	12/1998	De La Huerga	N/A	N/A
5884273	12/1998	Sattizahn et al.	N/A	N/A
5884457	12/1998	Ortiz et al.	N/A	N/A
5885245	12/1998	Lynch et al.	N/A	N/A
5891035	12/1998	Wood et al.	N/A	N/A
5891734	12/1998	Gill et al.	N/A	N/A
5893697	12/1998	Zimi et al.	N/A	N/A
5894273	12/1998	Meador et al.	N/A	N/A
5895371	12/1998	Levitas et al.	N/A	N/A
5897493	12/1998	Brown	N/A	N/A
5897530	12/1998	Jackson	N/A	N/A
5897989	12/1998	Beecham	N/A	N/A
5899665	12/1998	Makino et al.	N/A	N/A
5899855	12/1998	Brown	N/A	N/A
5901150	12/1998	Jhuboo et al.	N/A	N/A
5904668	12/1998	Hyman et al.	N/A	N/A
5905653	12/1998	Higham et al.	N/A	N/A
5907291	12/1998	Chen et al.	N/A	N/A
5907493	12/1998	Boyer et al.	N/A	N/A
5908027	12/1998	Butterfield et al.	N/A	N/A
5910107	12/1998	Iliff	N/A	N/A
5910252	12/1998	Truitt et al.	N/A	N/A
5911132	12/1998	Sloane	N/A	N/A
5911687	12/1998	Sato et al.	N/A	N/A
5912818	12/1998	McGrady et al.	N/A	N/A
5913197	12/1998	Kameda	N/A	N/A
5913310	12/1998	Brown	N/A	N/A
5915089	12/1998	Stevens et al.	N/A	N/A
5915240	12/1998	Karpf	N/A	N/A
5919154	12/1998	Toavs et al.	N/A	N/A
5921938	12/1998	Aoyama et al.	N/A	N/A
5923018	12/1998	Kameda et al.	N/A	N/A
5924074	12/1998	Evans	N/A	N/A
5924103	12/1998	Ahmed et al.	N/A	N/A
5927540	12/1998	Godlewski	N/A	N/A
5931791	12/1998	Saltzstein et al.	N/A	N/A
5935060	12/1998	Iliff	N/A	N/A
5935099	12/1998	Peterson et al.	N/A	N/A
5935106	12/1998	Olsen	N/A	N/A
5938413	12/1998	Makino et al.	N/A	N/A
5939326	12/1998	Chupp et al.	N/A	N/A
5939699	12/1998	Perttunen et al.	N/A	N/A
5940306	12/1998	Gardner et al.	N/A	N/A
5940802	12/1998	Hildebrand et al.	N/A	N/A
5941829	12/1998	Saltzstein et al.	N/A	N/A
5941846	12/1998	Duffy et al.	N/A	N/A
5942986	12/1998	Shabot et al.	N/A	N/A
5943423	12/1998	Muftic	N/A	N/A
5943633	12/1998	Wilson et al.	N/A	N/A
5944659	12/1998	Flach et al.	N/A	N/A
5945651	12/1998	Chorosinski et al.	N/A	N/A
5946083	12/1998	Melendez et al.	N/A	N/A
5946659	12/1998	Lancelot et al.	N/A	N/A
D414578	12/1998	Chen et al.	N/A	N/A
5950006	12/1998	Crater et al.	N/A	N/A
5951300	12/1998	Brown	N/A	N/A

5951510	12/1998	Barak	N/A	N/A
5954640	12/1998	Szabo	N/A	N/A
5954885	12/1998	Bollish et al.	N/A	N/A
5954971	12/1998	Pages et al.	N/A	N/A
5956023	12/1998	Lyle et al.	N/A	N/A
5957885	12/1998	Bollish et al.	N/A	N/A
5959529	12/1998	Kail, IV	N/A	N/A
5960085	12/1998	de la Huerga	N/A	N/A
5960403	12/1998	Brown	N/A	N/A
5960991	12/1998	Ophardt	N/A	N/A
5961446	12/1998	Beller et al.	N/A	N/A
5961448	12/1998	Swenson et al.	N/A	N/A
5961487	12/1998	Davis	N/A	N/A
5961923	12/1998	Nova et al.	N/A	N/A
5963641	12/1998	Crandall et al.	N/A	N/A
5964700	12/1998	Tallman et al.	N/A	N/A
5966304	12/1998	Cook et al.	N/A	N/A
5967975	12/1998	Ridgeway	N/A	N/A
5970423	12/1998	Langley et al.	N/A	N/A
5971593	12/1998	McGrady	N/A	N/A
5971921	12/1998	Timbel	N/A	N/A
5971948	12/1998	Pages et al.	N/A	N/A
5974124	12/1998	Schlueter, Jr. et al.	N/A	N/A
5975737	12/1998	Crater et al.	N/A	N/A
5980490	12/1998	Tsoukalis	N/A	N/A
5983193	12/1998	Heinonen et al.	N/A	N/A
5987519	12/1998	Peifer et al.	N/A	N/A
5991731	12/1998	Colon et al.	N/A	N/A
5993046	12/1998	McGrady et al.	N/A	N/A
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5995077	12/1998	Wilcox et al.	N/A	N/A
5995939	12/1998	Berman et al.	N/A	N/A
5995965	12/1998	Experton	N/A	N/A
5997167	12/1998	Crater et al.	N/A	N/A
5997476	12/1998	Brown	N/A	N/A
6003006	12/1998	Colella et al.	N/A	N/A
6004020	12/1998	Bartur	N/A	N/A
6004276	12/1998	Wright et al.	N/A	N/A
6006191	12/1998	DeRienzo	N/A	N/A
6006946	12/1998	Williams et al.	N/A	N/A
6009333	12/1998	Chaco	N/A	N/A
6010454	12/1999	Arieff et al.	N/A	N/A
6011858	12/1999	Stock et al.	N/A	N/A
6011999	12/1999	Holmes	N/A	N/A
6012034	12/1999	Hamparian et al.	N/A	N/A
6013057	12/1999	Danby et al.	N/A	N/A
6014631	12/1999	Teagarden et al.	N/A	N/A
6016444	12/1999	John	N/A	N/A
6017318	12/1999	Gauthier et al.	N/A	N/A
6018713	12/1999	Coli et al.	N/A	N/A
6019745	12/1999	Gray	N/A	N/A
6021392	12/1999	Lester et al.	N/A	N/A
6022315	12/1999	Iliff	N/A	N/A
6023522	12/1999	Draganoff et al.	N/A	N/A
6024539	12/1999	Blomquist	N/A	N/A
6024699	12/1999	Surwit et al.	N/A	N/A
6027217	12/1999	McClure et al.	N/A	N/A
6029138	12/1999	Khorasani et al.	N/A	N/A
6032119	12/1999	Brown et al.	N/A	N/A

6032155	12/1999	de la Huerga	N/A	N/A
6033076	12/1999	Braeuning et al.	N/A	N/A
6039251	12/1999	Holowko et al.	N/A	N/A
6039467	12/1999	Holmes	N/A	N/A
6047259	12/1999	Campbell et al.	N/A	N/A
6048086	12/1999	Valerino	N/A	N/A
6050940	12/1999	Braun et al.	N/A	N/A
6055487	12/1999	Margery et al.	N/A	N/A
6057758	12/1999	Dempsey et al.	N/A	N/A
6059736	12/1999	Tapper	N/A	N/A
6061603	12/1999	Papadopoulos et al.	N/A	N/A
6065819	12/1999	Holmes et al.	N/A	N/A
6068153	12/1999	Young et al.	N/A	N/A
6068156	12/1999	Liff et al.	N/A	N/A
6073046	12/1999	Patel et al.	N/A	N/A
6074345	12/1999	van Oostrom et al.	N/A	N/A
6079621	12/1999	Vardanyan et al.	N/A	N/A
6080106	12/1999	Lloyd et al.	N/A	N/A
6081048	12/1999	Bergmann et al.	N/A	N/A
6081786	12/1999	Barry et al.	N/A	N/A
6082776	12/1999	Feinberg	N/A	N/A
6083206	12/1999	Molko	N/A	N/A
6093146	12/1999	Filangeri	N/A	N/A
6095985	12/1999	Raymond et al.	N/A	N/A
6096561	12/1999	Tayi	N/A	N/A
6098892	12/1999	Peoples, Jr.	N/A	N/A
6101407	12/1999	Groezinger	N/A	N/A
6101478	12/1999	Brown	N/A	N/A
6102856	12/1999	Groff et al.	N/A	N/A
6108399	12/1999	Hernandez-Guerra et al.	N/A	N/A
6108588	12/1999	McGrady	N/A	N/A
6109774	12/1999	Holmes et al.	N/A	N/A
6112224	12/1999	Peifer et al.	N/A	N/A
RE36871	12/1999	Epstein et al.	N/A	N/A
6113554	12/1999	Gilcher et al.	N/A	N/A
6116461	12/1999	Broadfield et al.	N/A	N/A
6117940	12/1999	Mjalli	N/A	N/A
6123524	12/1999	Danby et al.	N/A	N/A
6125350	12/1999	Dirbas	N/A	N/A
6129517	12/1999	Danby et al.	N/A	N/A
6132371	12/1999	Dempsey et al.	N/A	N/A
6134504	12/1999	Douglas et al.	N/A	N/A
6135949	12/1999	Russo et al.	N/A	N/A
6139177	12/1999	Venkatraman et al.	N/A	N/A
6139495	12/1999	De La Huerga	N/A	N/A
6141412	12/1999	Smith et al.	N/A	N/A
6144922	12/1999	Douglas et al.	N/A	N/A
6145695	12/1999	Garrigues	N/A	N/A
6146523	12/1999	Kenley et al.	N/A	N/A
6148297	12/1999	Swor et al.	N/A	N/A
6149063	12/1999	Reynolds et al.	N/A	N/A
6151536	12/1999	Arnold et al.	N/A	N/A
6152364	12/1999	Schoonen et al.	N/A	N/A
6154668	12/1999	Pedersen et al.	N/A	N/A
6154726	12/1999	Rensimer et al.	N/A	N/A
6157914	12/1999	Seto et al.	N/A	N/A
6158965	12/1999	Butterfield et al.	N/A	N/A
6160478	12/1999	Jacobsen et al.	N/A	N/A
6161095	12/1999	Brown	N/A	N/A

6161141	12/1999	Dillon	N/A	N/A
6163737	12/1999	Fedor et al.	N/A	N/A
6165154	12/1999	Gray et al.	N/A	N/A
6168563	12/2000	Brown	N/A	N/A
6170007	12/2000	Venkatraman et al.	N/A	N/A
6170746	12/2000	Brook et al.	N/A	N/A
6171112	12/2000	Clark et al.	N/A	N/A
6171237	12/2000	Avitall et al.	N/A	N/A
6171264	12/2000	Bader	N/A	N/A
6173198	12/2000	Schulze et al.	N/A	N/A
6175779	12/2000	Barrett	N/A	N/A
6175977	12/2000	Schumacher et al.	N/A	N/A
6176392	12/2000	William et al.	N/A	N/A
6182047	12/2000	Dirbas	N/A	N/A
6183417	12/2000	Geheb et al.	N/A	N/A
6186145	12/2000	Brown	N/A	N/A
6192320	12/2000	Margrey et al.	N/A	N/A
6193480	12/2000	Butterfield	N/A	N/A
6195887	12/2000	Danby et al.	N/A	N/A
6198394	12/2000	Jacobsen et al.	N/A	N/A
6200264	12/2000	Satherley et al.	N/A	N/A
6200289	12/2000	Hochman et al.	N/A	N/A
6202923	12/2000	Boyer et al.	N/A	N/A
6203495	12/2000	Bardy	N/A	N/A
6203528	12/2000	Deckert et al.	N/A	N/A
6206238	12/2000	Ophardt	N/A	N/A
6206829	12/2000	Iliff	N/A	N/A
6210361	12/2000	Kamen et al.	N/A	N/A
6213391	12/2000	Lewis	N/A	N/A
6213738	12/2000	Danby et al.	N/A	N/A
6213972	12/2000	Butterfield et al.	N/A	N/A
6219439	12/2000	Burger	N/A	N/A
6219587	12/2000	Ahlin et al.	N/A	N/A
6221009	12/2000	Doi et al.	N/A	N/A
6221011	12/2000	Bardy	N/A	N/A
6221012	12/2000	Maschke et al.	N/A	N/A
6222619	12/2000	Herron et al.	N/A	N/A
6224549	12/2000	Drongelen	N/A	N/A
6225901	12/2000	Kail, IV	N/A	N/A
6226564	12/2000	Stuart	N/A	N/A
6226745	12/2000	Wiederhold	N/A	N/A
6230142	12/2000	Benigno et al.	N/A	N/A
6230927	12/2000	Schoonen et al.	N/A	N/A
6234997	12/2000	Kamen et al.	N/A	N/A
6245013	12/2000	Minoz et al.	N/A	N/A
6246473	12/2000	Smith, Jr. et al.	N/A	N/A
6248063	12/2000	Barnhill et al.	N/A	N/A
6248065	12/2000	Brown	N/A	N/A
6255951	12/2000	De La Huerga	N/A	N/A
6256643	12/2000	Cork et al.	N/A	N/A
6256967	12/2000	Hebron et al.	N/A	N/A
6259355	12/2000	Chaco et al.	N/A	N/A
6259654	12/2000	De La Huerga	N/A	N/A
6260021	12/2000	Wong et al.	N/A	N/A
6266645	12/2000	Simpson	N/A	N/A
6269340	12/2000	Ford et al.	N/A	N/A
D446854	12/2000	Cheney, II et al.	N/A	N/A
6270455	12/2000	Brown	N/A	N/A
6270457	12/2000	Bardy	N/A	N/A

6272394	12/2000	Lipps	N/A	N/A
6272505	12/2000	De La Huerga	N/A	N/A
6277072	12/2000	Bardy	N/A	N/A
6278999	12/2000	Knapp	N/A	N/A
6283322	12/2000	Liff et al.	N/A	N/A
6283944	12/2000	McMullen et al.	N/A	N/A
6290646	12/2000	Cosentino et al.	N/A	N/A
6290650	12/2000	Butterfield et al.	N/A	N/A
6294999	12/2000	Yarin et al.	N/A	N/A
6295506	12/2000	Heinonen	N/A	N/A
6304788	12/2000	Eady et al.	N/A	N/A
6306088	12/2000	Krausman et al.	N/A	N/A
6307956	12/2000	Black	N/A	N/A
6308171	12/2000	De La Huerga	N/A	N/A
6311163	12/2000	Sheehan et al.	N/A	N/A
6312227	12/2000	Davis	N/A	N/A
6312378	12/2000	Bardy	N/A	N/A
6314384	12/2000	Goetz	N/A	N/A
6317719	12/2000	Schrier et al.	N/A	N/A
6319200	12/2000	Lai et al.	N/A	N/A
6321203	12/2000	Kameda	N/A	N/A
6322502	12/2000	Schoenberg et al.	N/A	N/A
6322504	12/2000	Kirshner	N/A	N/A
6322515	12/2000	Goor et al.	N/A	N/A
6330491	12/2000	Lion	N/A	N/A
6332090	12/2000	DeFrank et al.	N/A	N/A
RE37531	12/2001	Chaco et al.	N/A	N/A
6337631	12/2001	Pai et al.	N/A	N/A
6338007	12/2001	Broadfield et al.	N/A	N/A
6339732	12/2001	Phoon et al.	N/A	N/A
6345260	12/2001	Cummings, Jr. et al.	N/A	N/A
6346886	12/2001	De La Huerga	N/A	N/A
6347553	12/2001	Morris et al.	N/A	N/A
6352200	12/2001	Schoonen et al.	N/A	N/A
6353817	12/2001	Jacobs et al.	N/A	N/A
6358225	12/2001	Butterfield	N/A	N/A
6358237	12/2001	Paukovits et al.	N/A	N/A
6361263	12/2001	Dewey et al.	N/A	N/A
6362591	12/2001	Moberg	N/A	N/A
6363282	12/2001	Nichols et al.	N/A	N/A
6363290	12/2001	Lyle et al.	N/A	N/A
6364834	12/2001	Reuss et al.	N/A	N/A
6368273	12/2001	Brown	N/A	N/A
6370841	12/2001	Chudy et al.	N/A	N/A
6381577	12/2001	Brown	N/A	N/A
6385505	12/2001	Lipps	N/A	N/A
6393369	12/2001	Carr	N/A	N/A
6397190	12/2001	Goetz	N/A	N/A
6401072	12/2001	Haudenschild et al.	N/A	N/A
6402702	12/2001	Gilcher et al.	N/A	N/A
6406426	12/2001	Reuss et al.	N/A	N/A
6407335	12/2001	Franklin-Lees et al.	N/A	N/A
6408330	12/2001	DeLaHuerga	N/A	N/A
6416471	12/2001	Kumar et al.	N/A	N/A
6421650	12/2001	Goetz et al.	N/A	N/A
6424996	12/2001	Killcommons et al.	N/A	N/A
6427088	12/2001	Bowman, IV et al.	N/A	N/A
6434531	12/2001	Lancelot et al.	N/A	N/A
6434569	12/2001	Tomlinson et al.	N/A	N/A

6438451	12/2001	Lion	N/A	N/A
RE37829	12/2001	Charhut et al.	N/A	N/A
6449927	12/2001	Hebron et al.	N/A	N/A
6450956	12/2001	Rappaport et al.	N/A	N/A
6458102	12/2001	Mann et al.	N/A	N/A
6461037	12/2001	O'Leary	N/A	N/A
6463310	12/2001	Swedlow et al.	N/A	N/A
6468242	12/2001	Wilson et al.	N/A	N/A
6470234	12/2001	McGrady	N/A	N/A
6471089	12/2001	Liff et al.	N/A	N/A
6471645	12/2001	Warkentin et al.	N/A	N/A
6471646	12/2001	Thede	N/A	N/A
6475146	12/2001	Frelburger et al.	N/A	N/A
6475148	12/2001	Jackson et al.	N/A	N/A
6475180	12/2001	Peterson et al.	N/A	N/A
6478737	12/2001	Bardy	N/A	N/A
6485465	12/2001	Moberg et al.	N/A	N/A
6494831	12/2001	Koritzinsky	N/A	N/A
6511138	12/2002	Gardner et al.	N/A	N/A
6519569	12/2002	White et al.	N/A	N/A
6537244	12/2002	Paukovits	N/A	N/A
6542902	12/2002	Dulong et al.	N/A	N/A
6542910	12/2002	Cork et al.	N/A	N/A
6544174	12/2002	West et al.	N/A	N/A
6544228	12/2002	Heitmeier	N/A	N/A
6551243	12/2002	Bocionek et al.	N/A	N/A
6551276	12/2002	Mann et al.	N/A	N/A
6554791	12/2002	Cartledge et al.	N/A	N/A
6554798	12/2002	Mann et al.	N/A	N/A
6555986	12/2002	Moberg	N/A	N/A
6558321	12/2002	Burd et al.	N/A	N/A
6561975	12/2002	Pool et al.	N/A	N/A
6562001	12/2002	Lebel et al.	N/A	N/A
6564104	12/2002	Nelson et al.	N/A	N/A
6564105	12/2002	Starkweather et al.	N/A	N/A
6564121	12/2002	Wallace et al.	N/A	N/A
6571128	12/2002	Lebel et al.	N/A	N/A
6575900	12/2002	Zweig et al.	N/A	N/A
6577899	12/2002	Lebel et al.	N/A	N/A
6579232	12/2002	Sakamaki et al.	N/A	N/A
6581069	12/2002	Robinson et al.	N/A	N/A
6581798	12/2002	Liff et al.	N/A	N/A
6584336	12/2002	Ali et al.	N/A	N/A
6585157	12/2002	Brandt et al.	N/A	N/A
6585644	12/2002	Lebel et al.	N/A	N/A
6585675	12/2002	O'Mahony et al.	N/A	N/A
6592551	12/2002	Cobb	N/A	N/A
6593528	12/2002	Franklin-Lees et al.	N/A	N/A
6602469	12/2002	Maus et al.	N/A	N/A
6607485	12/2002	Bardy	N/A	N/A
6610973	12/2002	Davis, III	N/A	N/A
6613009	12/2002	Bainbridge et al.	N/A	N/A
6616633	12/2002	Butterfield et al.	N/A	N/A
6635014	12/2002	Starkweather et al.	N/A	N/A
6648821	12/2002	Lebel et al.	N/A	N/A
6659948	12/2002	Lebel et al.	N/A	N/A
6664893	12/2002	Eveland et al.	N/A	N/A
6668196	12/2002	Villegas et al.	N/A	N/A
6669663	12/2002	Thompson	N/A	N/A

6673314	12/2003	Burbank et al.	N/A	N/A
6687546	12/2003	Lebel et al.	N/A	N/A
6689091	12/2003	Bui et al.	N/A	N/A
6694191	12/2003	Starkweather et al.	N/A	N/A
6694334	12/2003	DuLong et al.	N/A	N/A
6711460	12/2003	Reese	N/A	N/A
6731324	12/2003	Levy	N/A	N/A
6733447	12/2003	Lai et al.	N/A	N/A
6735497	12/2003	Wallace et al.	N/A	N/A
6740075	12/2003	Lebel et al.	N/A	N/A
6746398	12/2003	Hervy et al.	N/A	N/A
6758810	12/2003	Lebel et al.	N/A	N/A
6768425	12/2003	Flaherty et al.	N/A	N/A
6771369	12/2003	Rzasa et al.	N/A	N/A
6775602	12/2003	Gordon, Jr. et al.	N/A	N/A
6776304	12/2003	Liff et al.	N/A	N/A
6790198	12/2003	White et al.	N/A	N/A
6804656	12/2003	Rosenfeld et al.	N/A	N/A
6810290	12/2003	Lebel et al.	N/A	N/A
6811533	12/2003	Lebel et al.	N/A	N/A
6811534	12/2003	Bowman, IV et al.	N/A	N/A
6811707	12/2003	Rovatti et al.	N/A	N/A
6813473	12/2003	Bruker	N/A	N/A
6813519	12/2003	Lebel et al.	N/A	N/A
6814255	12/2003	Liff et al.	N/A	N/A
6820093	12/2003	De La Huerga	N/A	N/A
6842736	12/2004	Brzozowski	N/A	N/A
6847861	12/2004	Lunak et al.	N/A	N/A
6854088	12/2004	Massengale et al.	N/A	N/A
6871211	12/2004	Labounty et al.	N/A	N/A
6873268	12/2004	Lebel et al.	N/A	N/A
6877530	12/2004	Osborne et al.	N/A	N/A
6880034	12/2004	Manke et al.	N/A	N/A
6885288	12/2004	Pincus	N/A	N/A
6887201	12/2004	Bardy	N/A	N/A
6892941	12/2004	Rosenblum	N/A	N/A
6912549	12/2004	Rotter et al.	N/A	N/A
6913590	12/2004	Sorenson et al.	N/A	N/A
6915265	12/2004	Johnson	N/A	N/A
6915823	12/2004	Osborne et al.	N/A	N/A
6928452	12/2004	De La Huerga	N/A	N/A
6950708	12/2004	Bowman, IV et al.	N/A	N/A
6958705	12/2004	Lebel et al.	N/A	N/A
6974437	12/2004	Lebel et al.	N/A	N/A
6975924	12/2004	Kircher et al.	N/A	N/A
6976628	12/2004	Krupa	N/A	N/A
6979306	12/2004	Moll	N/A	N/A
6980958	12/2004	Surwit et al.	N/A	N/A
6981644	12/2005	Cheong et al.	N/A	N/A
6985870	12/2005	Martucci et al.	N/A	N/A
6991002	12/2005	Osborne et al.	N/A	N/A
6995664	12/2005	Darling	N/A	N/A
7006893	12/2005	Hart et al.	N/A	N/A
7015806	12/2005	Naidoo et al.	N/A	N/A
7017622	12/2005	Osborne et al.	N/A	N/A
7017623	12/2005	Tribble et al.	N/A	N/A
7028723	12/2005	Alouani et al.	N/A	N/A
7096212	12/2005	Tribble et al.	N/A	N/A
7117902	12/2005	Osborne	N/A	N/A

7151982	12/2005	Liff et al.	N/A	N/A
7194336	12/2006	DiGianfilippo et al.	N/A	N/A
7209891	12/2006	Addy et al.	N/A	N/A
7240699	12/2006	Osborne et al.	N/A	N/A
7255680	12/2006	Gharib	N/A	N/A
7277579	12/2006	Huang	N/A	N/A
7277757	12/2006	Casavant et al.	N/A	N/A
7317967	12/2007	DiGianfilippo et al.	N/A	N/A
7321861	12/2007	Oon	N/A	N/A
7343224	12/2007	DiGianfilippo et al.	N/A	N/A
7403901	12/2007	Carley et al.	N/A	N/A
7427002	12/2007	Liff et al.	N/A	N/A
7493263	12/2008	Helmus et al.	N/A	N/A
7499581	12/2008	Tribble et al.	N/A	N/A
7509280	12/2008	Haudenschild	N/A	N/A
7555557	12/2008	Bradley et al.	N/A	N/A
7561312	12/2008	Proudfoot et al.	N/A	N/A
7581953	12/2008	Lehmann et al.	N/A	N/A
7599516	12/2008	Limer et al.	N/A	N/A
7610115	12/2008	Rob et al.	N/A	N/A
7630908	12/2008	Amrien et al.	N/A	N/A
7636718	12/2008	Steen et al.	N/A	N/A
7672859	12/2009	Louie et al.	N/A	N/A
7698019	12/2009	Moncrief et al.	N/A	N/A
7698154	12/2009	Marchosky	N/A	N/A
7734478	12/2009	Goodall et al.	N/A	N/A
7753085	12/2009	Tribble et al.	N/A	N/A
7769601	12/2009	Bleser et al.	N/A	N/A
7783383	12/2009	Eliuk et al.	N/A	N/A
D624225	12/2009	Federico et al.	N/A	N/A
7801642	12/2009	Ansari et al.	N/A	N/A
7847970	12/2009	McGrady	N/A	N/A
7853621	12/2009	Guo	N/A	N/A
7904822	12/2010	Monteleone et al.	N/A	N/A
7931859	12/2010	Mlodzinski et al.	N/A	N/A
7937290	12/2010	Bahir	N/A	N/A
7986369	12/2010	Burns	N/A	N/A
7991507	12/2010	Liff et al.	N/A	N/A
8170271	12/2011	Chen	N/A	N/A
8191339	12/2011	Tribble et al.	N/A	N/A
8215557	12/2011	Reno et al.	N/A	N/A
8220503	12/2011	Tribble et al.	N/A	N/A
8225824	12/2011	Eliuk et al.	N/A	N/A
D667961	12/2011	Marmier	N/A	N/A
8267129	12/2011	Doherty et al.	N/A	N/A
8271138	12/2011	Eliuk et al.	N/A	N/A
8280549	12/2011	Liff et al.	N/A	N/A
8284305	12/2011	Newcomb et al.	N/A	N/A
8374887	12/2012	Alexander	N/A	N/A
8386070	12/2012	Eliuk et al.	N/A	N/A
8548824	12/2012	daCosta	N/A	N/A
8554579	12/2012	Tribble et al.	N/A	N/A
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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims priority from U.S. Provisional Application No. 62/068,301 filed on Oct. 24, 2014, entitled “AUTOMATED EXCHANGE OF HEALTHCARE INFORMATION FOR FULFILLMENT OF MEDICATION DOSES,” the contents of which are incorporated by reference herein as if set forth in full.

BACKGROUND
(1) Despite advances in electronic healthcare information systems, processes related to the exchange of information may still rely upon manual intervention by a human user. While independent systems may be operative to maintain an electronic medical record (EMR), the exchange of information between independent systems may be complicated by a lack of a standardized data format or data exchange protocol. Furthermore, to the extent data may be exchanged between independent systems, use of unique semantics at each independent system may result in complications when exchanging data therebetween.
(2) For example, in the context of dose order processing, a physician or the like may rely upon a hardcopy order form to request a medication order to be administered to a patient. In turn, the hardcopy of the order form may provided to a pharmacy by way of, for example, physical transport, facsimile, relay by telephone, relay by email, etc. In turn, upon receipt of the information related to the dose order at the pharmacy, a technician may be required to manually input the information (e.g., by transcribing information from the hardcopy or an electronic copy thereof) regarding the requested dose order. As such, a new instance of an EMR related to the dose order requested based on the transcribed dose order may be generated in the pharmacy. In turn, the dose order may be fulfilled by the pharmacy (e.g., including calculation, review, compounding, and delivery of the dose order).
(3) As such, at one or more instances during the dose ordering process, information related to the requested dose order may be required to be manually processed by a human user. For example, a human user may be required to relay information from a physical hardcopy order form to a pharmacy system. Furthermore, a human user may be required to retrieve information from a first system and transcribe information into a second system. For example, a human user may be required to input received order information into a pharmacy information system (PIS). In this regard, despite advances in EMR systems for dose order generation and pharmacy fulfillment, the exchange of healthcare information may still be subject to human user handling.
(4) In turn, the potential for delays and errors may be compounded because of the requirement for human intervention. For example, it is been found that 60% of clinicians working with parenteral nutrition orders report one to five ordering errors per month. Significantly, near fatal complication or death may result in 4.8% of parenteral nutrition order errors in one two year span that was reviewed. Further still, transcription may provide delay in order fulfillment as it has been found that up to 10% of parenteral nutrition orders require clarification. The most common causes of the need for clarification include illegible writing, missing essential ingredients, or unstable macronutrient content. In this regard, human errors may come in the form of transcription errors, EMR order entry errors, errors resulting in information asymmetry, errors resulting in order form legibility, errors related to reentry of orders in the pharmacy system, or other points within a data flow that require human intervention. Accordingly, improvements in exchange of healthcare information between independent healthcare systems may reduce the reliance upon human intervention, thereby reducing error rates in improving patient outcomes.
SUMMARY
(5) In view the foregoing, the present disclosure is generally related to improved exchange of healthcare information. Specifically, the present disclosure presents systems and methods that may at least in part help reduce reliance upon manual human interaction in the exchange of healthcare information between independent healthcare systems. In turn, the reliance upon human interaction to facilitate information exchange may be reduced, and the speed and accuracy at which healthcare information may be exchange between independent healthcare systems may be improved. In turn, an at least partially automated approach to healthcare information exchange may be facilitated.
(6) Accordingly, use of an automated health information exchange as described herein may improve patient outcomes by assisting in reduction of errors associated with human interaction in the exchange of healthcare information. Furthermore, the speed at which information may be exchanged may be increased for more efficient pharmacy

operation. For instance, one context in which automated healthcare information exchange may be utilized is in the context of dose order fulfillment. Specifically, automation of the exchange of healthcare information from dose order entry to dose order fulfillment may be facilitated to assist in reduction of (and potentially elimination of) the need for human intervention in connection with the exchange of healthcare information between an EMR system that facilitates dose order entry, pharmacy information systems (PIS) for management of pharmacy operations, and dose fulfillment clients for use in fulfillment of medication doses that are to be administered to a patient. As discussed herein, dose orders may relate to requested medication doses in any number of contexts including, for example, IV doses, parenteral nutrition (PN) doses, total parenteral nutrition (TPN) doses, or the like.

(7) Specifically, the disclosure presented herein may facilitate receipt and processing of a healthcare information data stream by a platform interface module. The platform interface module may perform identification and parsing of the healthcare information data stream to identify individual dose orders from the healthcare information data stream. In this regard, the healthcare information data stream may include a multitude of data including, for example, a plurality of dose orders or other data related to EMR data or hospital information system (HIS) data. Upon identification of a dose order from the healthcare information data stream, the platform interface module may be operative to parse the dose order such that dose order data fields may be identified and populated with dose order metadata related to the dose order. The dose order data fields may correspond to dose order characteristics such as an ingredient list, product requirements, administration characteristics, patient information, or other appropriate data types related to the dose order. In turn, corresponding dose order metadata for each respective dose order may be parsed by the platform interface module. A staging table may be generated and stored at the platform interface module in relation to a dose order. The staging table may include a standardized intermediate format for storage of dose order metadata in corresponding relation to dose order data fields for a given dose order.

(8) In this regard, it may be appreciated that different various EMR systems (e.g., one or more HISs, one or more physician order entry (POE) systems, or the like) may be used to generate and/or transmit dose orders to a pharmacy. In turn, receipt and processing of the health information data stream to generate staging tables may facilitate generation of a standardized intermediate format for the dose order. The standardized intermediate format may be beneficial as the dose order may be routed to one of a plurality of different dose order fulfillment clients for use in fulfillment of the medication dose associated with the dose order. In turn, rather than having to facilitate specific respective correlations between different EMR system formats and specific dose fulfillment client input formats, the standardized intermediate form (e.g., the staging table) may be utilized such that a plurality of different EMR system formats may be standardized into the standardized intermediate form and the standardized intermediate form may be in turn transformed into one of the plurality of different input formats associated with different respective ones of dose fulfillment clients. In this regard, upon addition of different EMR systems, the data from the EMR system may be standardized into the standardized intermediate format such that the system may process dose orders received from the new EMR system without having to generate application functionality specific to each dose fulfillment client for the new EMR system. Furthermore, as new dose fulfillment clients are added the system, transformation of the data in the standardized intermediate format may be accomplished without having to generate application functionality for the new dose fulfillment client specific to each EMR system. Accordingly, a robust and modular data exchange system may be realized that allows for any one of a plurality of EMR systems to be utilized in conjunction with any one of a plurality of dose fulfillment clients.

(9) In at least some embodiments, the dose order records maintained in a staging table may have some dose order management functionality applied thereto. For example, functions in connection with viewing, modifying, prioritizing, or otherwise organizing the dose orders may be applied to the dose order information stored in the staging tables. In turn, improved pharmacy management may be provided in that personnel associated with the operation of the pharmacy may be able to perform pharmacy management functions collectively on the dose orders stored in the staging tables. For instance, such pharmacy management may be performed on the dose orders prior to provision of the dose orders to a plurality of dose fulfillment clients. Further still, dose order management may be performed locally relative to one or more dose fulfillment clients. In turn, pharmacy resources may be efficiently managed in connection with dose orders having different priorities, urgencies, or the like.

(10) Dose order information from the staging table may be retrieved by and/or provided to a dose fulfillment client for use in preparation of a dose associated with the dose order. In this regard, the dose order information in the staging table may be transformed from the standardized intermediate form of the staging table into an input format associated with a dose fulfillment client. For instance, a unique input format for a given dose fulfillment client may be provided. As such, identification of a specific dose fulfillment client for use in preparation of the dose order may allow for the corresponding unique input format associated with the identified dose fulfillment client to be used to transform the data from the staging table into the unique input format for a given client.

(11) In connection with the transformation of information from the staging table into the dose from a client specific format, a mapping and/or translation processes may occur. For example, the transformation of the dose order information from the standardized intermediate format to the unique input format for a given dose fulfillment client may include mapping dose order data fields from the staging table to corresponding respective dose fulfillment client

input fields. In addition to the specific mapping, translations of the data may be performed. For example, the unique input format for a dose fulfillment client may specify a form in which the data is expected. As such, data may be translated from the form in which it is provided in the staging table such that the data complies with the form of the unique input format.

(12) Additionally, dose order metadata related to the corresponding respective dose fulfillment client input fields may be validated to determine whether a valid input is provided for the dose order from the staging table. In the event the data may not be validated (e.g., due to a parsing error, a mapping error, a translation error, or some other error), exception processing may occur. That is, exception processing may allow for a user to resolve any errors identified during validation, mapping, or translation. The resolution of the errors by a user may be used to modify processing rules (e.g., validation rules, mapping rules, or translation rules) or the like for subsequent dose orders. The receipt of an input corresponding to a resolution of an error by a user may be incorporated into the transformation and/or validation of subsequent orders. Thus, in connection with the transformation process, exceptions related to the mapping, translation, and/or validation of dose order metadata from the staging table may be processed as will be discussed herein. Furthermore, logistical processing may be performed such that changes, cancellations, or other modifications to a dose order received during the processing of the dose order may be facilitated. The exemption and/or logistical processing may be performed at the same location or different locations. Furthermore, the exemption and/or logistical processing may occur at the platform interface module, transformation module, and/or dose fulfillment client.

(13) In addition, logging may occur throughout the exchange of healthcare data between independent systems described herein. In turn, auditing, troubleshooting, or the like may be facilitated for any or all of the steps of the exchange between systems described herein. The log information generated during the healthcare information exchange may be maintained at each independent system or aggregated into a single repository related to a portion of or the entire path of the data exchange. In turn, improve record-keeping for the purposes of auditing or the like may provided in connection with the exchange of healthcare information between systems described herein.

(14) As contemplated herein, a plurality of dose fulfillment clients may be employed in connection with the healthcare information exchange facilitated by the disclosure presented herein. For instance, dose fulfillment clients may include a pharmacy workflow management application for use in manual preparation of dose orders, automated syringe filling platforms, TPN management modules, automated compounders (e.g., for use in fulfilling PN and TPN orders), automated dose dispensing cabinets, or other dose fulfillment clients that may be utilized to realize a physical dose associated with the dose order. Accordingly, at least some of the dose fulfillment clients may comprise automated dose preparation systems for fulfillment of dose orders without the need for human intervention. In turn, in at least some embodiments facilitated herein, the processing performed on the dose order may be completely automated such that there is no human intervention between the entry of the dose order by a physician or the like and the fulfillment of the dose order by an automated dose fulfillment client.

(15) A first aspect of the disclosure presented herein includes a method for automated transformation of healthcare information by a transformation module between a first form from an electronic medical record (EMR) system and a second form associated with a dose fulfillment client for preparation of a dose in accord with a dose order of the health information data. The method includes retrieving dose order metadata, at a transformation module, for a dose order from a staging table corresponding to the dose order. The staging table is populated with dose order metadata received from a healthcare information data stream in a first form from the EMR system. Additionally, the staging table includes a plurality of dose order data fields populated with corresponding respective portions of the dose order metadata. The method further includes transforming the dose order metadata into a predefined second form corresponding with a dose fulfillment client and providing the dose order metadata in the second form to the dose fulfillment client for fulfillment of a dose associated with the dose order based on the dose order metadata.

(16) A number of feature refinements and additional features are applicable to the first aspect. These feature refinements and additional features may be used individually or in any combination. As such, each of the following features that will be discussed may be, but are not required to be, used with any other feature or combination of features of the first aspect.

(17) For example, the method may further include receiving the healthcare information data stream at a platform interface module from the EMR system. The healthcare information data stream may include information related to the dose order (e.g., among other information related to different dose orders and/or unrelated data). The method may also include identifying the dose order from the healthcare information data stream and parsing the dose order metadata for the dose order from the healthcare information data stream. The dose order metadata may include one or more dose characteristics associated with the dose order. The method may further include populating the staging table, stored in a staging table database at the platform interface module, with the dose order metadata for the dose order.

(18) The staging table may store the dose order metadata a standardized intermediate format. For example, the method may further include receiving a first dose order in a first EMR form and receiving a second dose order in a second EMR form. The first dose order and the second dose order may correspond to a dose order with identical

constituent ingredients. The form of the constituent ingredients in the first EMR form may differ from the second EMR form (e.g., because of different EMR sources or different EMR formats utilized). The form of the constituent ingredients for the first dose order may, in turn, be identical to the form of the constituent ingredients for the second dose order in the respective staging tables corresponding to the first dose order and the second dose order. Furthermore, other dose data may be standardized such as, for example, patient data, dose administration data, or the like.

(19) The method may also include identifying, at least in part based on the dose order metadata, the dose fulfillment client from a plurality of dose fulfillment clients for fulfillment of the dose order. As such, the plurality of dose fulfillment clients each have respective predefined second forms. In turn, the transforming may at least in part be based on the respective predefined second form of the identified dose fulfillment client. The staging table may be independent of any of the plurality of dose fulfillment clients.

(20) The transforming may include mapping the plurality of dose order data fields of the staging table to corresponding respective ones of a plurality of dose fulfillment client input fields. In an application, the dose fulfillment client input fields may be defined by a unique input format associated with the dose fulfillment client. The transforming may additionally include validating the dose order metadata of the plurality of dose order data fields with respect to the unique input format for corresponding ones of the dose fulfillment client input fields. The validating may include correlating the dose order metadata of the plurality of dose order data fields to formulary records of the dose fulfillment client with respect to the corresponding respective ones of the dose fulfillment client input fields.

(21) In an embodiment, the method may include exception processing (e.g., in response to the validating, mapping, or translating of dose order data). In this regard, the method may include generating an exception in response to an error associated with at least one of the mapping or validating. Additionally, the exception may include prompting a human user to resolve the error. As such, the method may include receiving, from the human user, an input associated with the resolution of the error. In turn, the input may include at least one of a correct mapping between a dose order data field of the staging table and a corresponding respective one of a dose fulfillment client input fields or a correct correlation between a portion of dose order metadata and a formulary record of the dose fulfillment client. As such, the exception processing may further include updating at least one of a mapping logic or a correlation logic for use in the mapping and correlating, respectively, based on the input received from the human user.

(22) In an embodiment, the method may include updating the staging table to indicate the dose order metadata for the dose order has been retrieved by the dose fulfillment client. As such, the processing status of a dose may be maintained at the staging table to provide an indication of dose orders that have been processed and those that have not been processed. The method may also include managing the dose order (e.g., by organizing, prioritizing, etc.) prior to providing the dose order metadata in the second form to the dose fulfillment client. For instance, the managing may include providing a user interface to allow a user to perform a management function relative to the dose order. Specifically, the managing may include at least one of modification of dose order metadata, cancellation of the dose order, organization of the dose order relative to other dose orders, or prioritization of the dose order relative to other dose orders.

(23) Additionally, the method may include performing logistical processing on the dose order. The logistical processing may include performing an action relative to the dose order in response to receipt of an EMR system message subsequent to the receipt of the dose order. The EMR system message may include at least one of a dose order change message or a dose order discontinuation message. As such, the logistical processing may include determining if the dose order metadata has been provided to the dose fulfillment client, and wherein the logistical processing is at least in part based on whether the determining. The logistical processing may include performing an action relative to the dose order record corresponding to the EMR system message for a dose order record that has not yet been provided to the dose fulfillment client. For instance, the logistical processing may include providing data to the dose fulfillment client related to the EMR system message for a dose order record that has been provided to the dose fulfillment client. The data provided to the dose fulfillment client related to the EMR system message may also be transformed into a form corresponding to the dose fulfillment client.

(24) In an embodiment, the dose fulfillment client may include at least one of a pharmacy workflow management application for manual preparation of the dose order, an automated dose preparation device, an automated total parenteral nutrition (TPN) compounder, or a dose dispensing cabinet. Additionally, the EMR system may include a hospital information system (HIS). As such, the method may include exchange of a message from independent healthcare systems comprising the EMR system and the dose fulfillment client. The data exchanged between the EMR system and the dose fulfillment client may be otherwise incompatible without the use of the healthcare information exchange system. For instance, the healthcare information data stream may be a Health Level 7 (HL7) format.

(25) The method may also include logging activity taken with respect to the dose order to generate a log regarding activities relative to the dose order. The logging may include aggregating a plurality of logs generated relative to different respective actions taken with respect to the dose order.

(26) A second aspect includes a method for automated exchange of healthcare information between an electronic medical record (EMR) system and a dose fulfillment client for fulfillment of a dose order. The method includes receiving a healthcare information data stream at a platform interface module. The healthcare information data stream includes information related to a dose order. The method also includes identifying the dose order from the healthcare information data stream and parsing dose order metadata for the dose order from the healthcare information data stream. The dose order metadata includes one or more dose characteristics associated with the dose order. The method further includes populating a staging table with the dose order metadata for the dose order. The staging table includes a plurality of dose order data fields populated with corresponding respective portions of the dose order metadata. The method further includes transforming the dose order metadata into a predefined format corresponding to a particular dose fulfillment client to generate transformed dose order metadata. Also, the method includes providing the particular dose fulfillment client access to the transformed dose order metadata, wherein the dose fulfillment client is operable to retrieve the transformed dose order metadata from the staging table to fulfill the dose order based on the transformed dose order metadata.
(27) A number of feature refinements and additional features are applicable to the second aspect. These feature refinements and additional features may be used individually or in any combination. As such, each of the following features that will be discussed may be, but are not required to be, used with any other feature or combination of features of the second aspect. For instance, any of the foregoing features described in relation to the first aspect may be applicable to the second aspect.
(28) A third aspect includes a healthcare information exchange system for exchange of healthcare information between an electronic medical record (EMR) system and a dose fulfillment client for fulfillment of a dose order. The system includes a stream processing module in operative communication with an EMR system to receive a healthcare information data stream from the EMR system. The system further includes a staging table database in operative communication with the stream processing module for storage of a staging table populated with dose order metadata included in the healthcare information data stream. The staging table includes a plurality of dose order data fields populated with corresponding respective portions of the dose order metadata. The system further includes a transformation module operatively disposed between the staging table data base and a dose fulfillment client. The transformation module is operative to identify the dose fulfillment client for use in fulfillment of the dose order and transform the dose order metadata into a predefined format corresponding to the dose fulfillment client. Specifically, the predefined format defines dose order data fields and corresponding dose order metadata formats associated with the dose fulfillment client.
(29) A number of feature refinements and additional features are applicable to the third aspect. These feature refinements and additional features may be used individually or in any combination. As such, each of the following features that will be discussed may be, but are not required to be, used with any other feature or combination of features of the third aspect. For instance, any of the foregoing features described in relation to the first aspect may be applicable to the third aspect.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a schematic view of an embodiment of a system for facilitating exchange of healthcare information.
(2) FIG. **2** is a flowchart depicting an embodiment of a method for exchange of healthcare information.
(3) FIG. **3** is a schematic view of an embodiment of a system for facilitating exchange of healthcare information between an EMR system and a plurality of dose fulfillment clients.
(4) FIG. **4** is a schematic view of an embodiment of a system for facilitating exchange of healthcare information between an EMR system and a dose fulfillment client for fulfillment of total parenteral nutrition dose.
(5) FIG. **5** is a schematic view of an embodiment of a system for facilitating exchange of healthcare information between an EMR system and a pharmacy workflow management application for fulfillment of a dose order.
(6) FIGS. **6-8** are schematic views of embodiments of a system for facilitating exchange of healthcare information between an EMR system and a plurality of dose fulfillment clients.
(7) FIG. **9** is an embodiment of a textual representation of a mapping between a healthcare information data stream and an input format for a dose fulfillment client.
(8) FIG. **10** is an embodiment of a human readable representation of an EMR system message comprising a dose order.
(9) FIG. **11** is an embodiment of a human readable representation of the operation of a transformation module executing relative to the EMR system message of FIG. **10**.
(10) FIG. **12** is an embodiment of a dose fulfillment client interface corresponding to the dose order of the EMR system message of FIG. **10**.
(11) FIG. **13** is an embodiment of an interface for display of dose orders processed by a healthcare information

exchange system.
(12) FIGS. **14-16** depict a sequence of interfaces related to an embodiment of exception handling of a dose order with processing errors in relation to the exchange of data related to the dose order.
(13) FIG. **17** is an embodiment of an interface for management of a transformation of healthcare information in a healthcare information exchange system.
(14) FIG. **18** is an embodiment of an interface for a platform interface module depicting standardization of an EMR message into an intermediate standardized format.

DETAILED DESCRIPTION

(15) While the invention is susceptible to various modifications and alternative forms, specific embodiments thereof have been shown by way of example in the drawings and are herein described in detail. It should be understood, however, that it is not intended to limit the invention to the particular form disclosed, but rather, the invention is to cover all modifications, equivalents, and alternatives falling within the scope of the invention as defined by the claims.
(16) FIG. **1** schematically depicts an embodiment of a system **100** for automated exchange of healthcare information. The system **100** may include an EMR system **110** that is in operative communication with a platform interface module **120**. In this regard, the EMR system **110** may provide a healthcare information data stream **105** to the platform interface module **120**. While a single EMR system **110** is shown in FIG. **1**, it may be appreciated that a plurality of EMR systems **110** may be in operative communication with the platform interface module **120**. The EMR systems **110** may correspond to different EMR systems **110** at a given facility or EMR systems **110** from a plurality of facilities. In any regard, the platform interface module **120** may be in operative communication with a transformation module **130**. The transformation module **130** may in turn be in operative communication with a dose fulfillment client **140**. As discussed in greater detail below, the transformation module **130** may be in operative communication with a plurality of dose fulfillment clients **140**. As such, the transformation module **130** may include a dose routing module and/or perform dose routing functions in relation to dose orders received at the transformation module **130**.
(17) In this regard, the system **100** in FIG. **1** may be operative to facilitate automated data exchange between independent healthcare information systems. For example, the EMR system **110** and the dose fulfillment client **140** may each correspond to independent healthcare information systems and/or products. As described above, the EMR system **110** and the fulfillment client **140** may use unrelated or incompatible data formats. In turn, the platform interface module **120** and transformation module **130** may act upon the health information data stream **105** to transform the data from the form received from the EMR system **110** such that the healthcare information (e.g., comprising dose order data) may be exchanged between the EMR system **110** in the dose fulfillment client **140**. Specifically, the data may be exchanged between the EMR system **110** and the dose fulfillment client **140** in an automated manner that avoids the requirement for human intervention with respect to the exchange of the healthcare information. In this regard and as described above, the automation of the exchange of information between the EMR system **110** and the dose fulfillment client **140** provided according to the disclosure presented herein may assist in reduction of errors and/or delays associated with human involvement with the healthcare information data stream **105** as it passes from the EMR system **110** to the dose fulfillment client **140**.
(18) The EMR system **110** may comprise a hospital information system (HIS). Accordingly, the healthcare information data stream **105** may comprise information regarding dose orders to be prepared by the dose of fulfillment client **140**. However, the healthcare information data stream **105** may also include other healthcare information that is unrelated to dose orders or preparation of dose orders. For example, in the case of an HIS, the healthcare information data stream **105** may include patient related data (e.g., admission/discharge data), hospital management data (e.g., resource availability, scheduling, etc.), billing and accounting data, or other ancillary data unrelated to dose orders or the preparation of dose orders.
(19) The healthcare information data stream **105** may be provided in a number of different formats. Furthermore, a given one platform interface module **120** may interface with a plurality of different ERM systems **110** each potentially providing healthcare information data streams **105** in differing formats. In an embodiment, the healthcare information data stream **105** may comprise a Health Level 7 (HL7) format. The HL7 format refers to set of loosely standardized conventions for the transfer of clinical and administration data between HISs. Despite the loose standardization of conventions provided by HL7 messages, the exchange of healthcare information between various independent healthcare systems employing HL7 formatting is not seamless. In this regard, different systems may utilize different lexicons, conventions, semantics, or other different standards within the HL7 format such that the exchange of information between two systems utilizing an HL7 format may still require intervention to allow for data exchange between systems.
(20) Accordingly, the platform interface module **120** may receive the healthcare information data stream **105** from the EMR system **110** regardless of the nature of the format used (e.g., regardless of variations in HL7 formats). For instance, different specific EMR systems **110** (e.g., different HISs) may utilized varying HL7 formats, all of which may be received by the platform interface module **120**. As will be discussed in greater detail below, the platform

interface module **120** may be operative to identify and/or parse dose orders from the healthcare information data stream **105**. The platform interface module **120** may in turn store the dose order records identified and parsed from the healthcare information data stream **105** in an a standardized intermediate format. For example, the standardized intermediate format may comprise a staging table for each respective dose order for which dose order metadata is parsed from the healthcare information data stream **105**. In turn, the parsed dose order metadata may be used to populate fields of the staging table that correspond to dose order data fields from the healthcare information data stream **105**.

(21) It should be noted that the standardized intermediate format (i.e., the staging table) may standardized relative to different HL7 formats used by different HL7 providers. That is, if first HL7 provider provides a dose order with identical constituent ingredients as a second HL7 provider, the portions of the HL7 messages corresponding to the identical constituent ingredients for the doses may still differ based on differing usage of the HL7 format by the first and second providers to describe the identical constituent ingredients. However, the platform interface module **120** may standardize the differing portions of the HL7 messages regarding the constituent ingredients for the identical orders such that the constituent ingredients for the orders would appear identical in the standardized intermediate format. Furthermore, patient data, dose administration data, or other data related to the dose may be provided in different formats corresponding to different uses of the HL7 format. However, this data may also be standardized in the standardized intermediate format. In this regard, the staging table comprising the standardized intermediate format may standardize HL7 messages received in different healthcare data streams **105**.

(22) The generation of a standardized intermediate format is illustrated in FIG. **18** which depicts a user interface **1800** and may display a listing **1805** of receive dose orders. A user may be able to select a selected one **1810** of the dose orders from the listing **1805** of dose orders. Upon selection of the selected one **1810**, corresponding information regarding the selected dose order **1810** may be displayed in an error pane **1820**, a pre-standardization pane **1830**, and a post-standardization pane **1840**. In this regard, the error pane **1820** may provide information to a user regarding errors (e.g., parsing errors, identification errors, or other errors that may occur with respect to receipt of the dose order from an EMR system **110**). Furthermore, the pre-standardization pane **1830** may display the dose order record as received from the EMR system **110**. In this regard, the pre-standardization pane **1830** may display a textual representation of an HL7 message received from the EMR system **110** prior to undergoing standardization for population of a staging table. The post-standardization pane **1840** may display the corresponding textual representation of the dose order after having been standardized into the standardized intermediate format. That is, the post standardization pane **1840** may provide a standardized version of the pre-standardized HL7 format displayed in the pre-standardization pane **1830**. Accordingly, the user interface **1800** may allow a user to review errors related to a selected dose order **1810**. Furthermore, a user may be operative to review pre-standardization and post-standardization formats side-by-side for a given dose order to facilitate review and or troubleshooting related to the dose order.

(23) The use of a standardized intermediate format may be particularly beneficial in the context of multiple healthcare data stream providers (e.g., multiple EMR systems **110** that may include, for example, HISs, PISs, POEs, or other healthcare information data stream **105** sources) and multiple dose fulfillment clients **140**. For instance, rather than having to develop specific transformation logic for all possible combinations of EMR systems **110** and all dose fulfillment clients **140**, the system **100** may provide a more modular approach. That is, as new EMR systems **110** are added to the system **100**, a parsing module (described in greater detail below) may be provided to format the healthcare information data stream **105** into the standardized intermediate format for storage in a staging table. In turn, all existing dose fulfillment clients **140** may receive information as a transformation module **130** may be capable of transforming the standardized intermediate format into each dose fulfillment client **140** specific format. Further still, as different dose fulfillment clients **140** are added to the system that require unique input formats, rather than having to develop transformation logic for each potential EMR system **110**, transformation logic relative to the standardized intermediate format may be generated. Accordingly, the standardized intermediate format allows for improved, modular development as additional components to which data is to be exchanged are added to the system **100**.

(24) The transformation module **130** may be operative to access the data regarding dose orders from the staging table maintained at the platform interface module **120**. The transformation module **130** may be operative to transform the accessed data in the standardized intermediate format regarding the dose orders from the standardized intermediate format of the staging table into a specific format association with a dose fulfillment client. In this regard, the client-specific format may correspond to a unique input format associated with and/or required by a dose fulfillment client **140**. In this regard, the unique input format for a given dose fulfillment client **140** may have input fields corresponding to dose order data fields of the staging table. In this regard, the transformation module **130** may map dose order data fields to corresponding input fields for the unique input format for a given dose fulfillment client **140**. Furthermore, the transformation client **130** may translate dose order metadata from a first format stored in the staging table into a target format associated with the unique input format for a given dose fulfillment client **140**. In this regard, the transformation module **130** may be operative to apply a transformation that is specific to a dose

fulfillment client **140** to dose order data stored in the staging table based on a dose fulfillment client **140** that is identified to be used to fulfill the corresponding dose order.

(25) As such, data regarding a dose order may be provided to a dose fulfillment client **140** in a format associated with a unique input format associated with the dose fulfillment client **140** such that the dose order data may be automatically provided to the dose fulfillment client **140** for use by the dose fulfillment client **140** in fulfillment of the dose corresponding to the dose order data. As will be appreciated from the further discussion below, the dose fulfillment client **140** may comprise any one or more of a number of different types of dose fulfillment clients that may be utilized in the fulfillment of a dose associated with a dose order. For example, the dose fulfillment client **140** may comprise any one or more of a pharmacy workflow management application for management of manual preparation of the dose, a total parenteral nutrition (TPN) client for processing in connection with and/or preparation (e.g., automated preparation) of a TPN dose order, an automated syringe filler, a dose dispensation cabinet with pre-stocked doses that may be allocated based on received dose order data to fulfill a dose, etc.

(26) As will also be appreciated from the discussion below, the platform interface module **120** and transformation module **130** may each include hardware and/or software that comprise each respective module. Furthermore, the platform interface module **120** and the transformation client **130** may be collectively provided as a single module. That is, each module may comprise hardware and/or software for execution of functionality described below in relation to each respective one of the platform interface module **120** and/or transformation module **130**. For example, the respective modules may individually or collectively comprise one or more processors in operative communication with a memory device that stores non-transitory machine-readable data that may be used to specifically configure the one or more processor for execution of functionality related to the modules as described below. In this regard, the respective modules may also comprise the memory device that stores non-transitory machine-readable data structure (e.g., such as a hard drive, flash drive, memory module, or other appropriate physical electronic memory) for storage of the non-transitory machine-readable data used for specific configuration of the one or more processors.

(27) With further reference to FIG. **2**, an embodiment of a method **200** for automated exchange of healthcare information between an EMR system **110** and a dose fulfillment client **140** is shown as a flowchart. While the method **200** is described herein in connection with the system **100** of FIG. **1**, it will further be appreciated that the method **200** may be performed in connection with any other embodiment presented below. Furthermore, while the method **200** is depicted in the form of a sequential flow chart with specific discrete operations in a particular order, it is contemplated that the steps of the method discussed below may be performed in any order (or concurrently) unless otherwise specified. Also, further embodiments of the method are contemplated wherein some steps may be omitted. That is, all steps of the process need not be performed in each embodiment.

(28) The method **200** may include receiving **210** a dose order data stream **105** (e.g., such as a health information data stream **105** comprising one or more dose orders therein). As may be appreciated, the receiving **210** may include transmission of the dose order data stream from the EMR system **110**. In this regard, the dose order data stream may be sent upon initiation from the EMR system **110** (i.e., “pushed” from the EMR system **110**) and/or be sent upon request by the platform interface module **120** (i.e., “pulled” from the EMR system **110**). The method **200** may also include identifying/parsing **215** dose order metadata from the dose order data stream. In this regard, the identifying/parsing **215** may include analyzing the dose order data stream **105** to identify individual dose orders from the data stream **105**. For example and as described above, the data stream **105** may include a plurality of dose orders and/or data that is unrelated to dose orders at all. In turn, the identifying may include delineating individual dose orders from the data stream **105**. As such, the identifying may include recognition of data in a specific format indicative of a dose order. Additionally or alternatively, the data stream **105** may include express delineators to assist in identifying individual ones of the dose orders in the data stream **105**. Additionally, dose order metadata may be parsed from the data stream **105** in regard an individual dose order. In this regard, the parsing may include locating one or more specific dose order data fields and corresponding dose order metadata related to the data fields. In turn, the dose order data fields and dose order metadata related thereto may be used in populating a dose order data field. In addition, standardization of the dose order metadata may be performed such that the staging table comprising the dose order data fields may include standardized dose order metadata (i.e., in a standardized intermediate format).

(29) Any one or more of a plurality of dose order data fields may be included in the data stream **105** and, in turn, in a corresponding staging table related to the identified/parsed dose order may be provided and/or generated. As an example, a predefined staging table may be provided that includes predefined data fields that relate to patient specific data, ingredients of the dose order, administration details for the dose order, or other appropriate data fields that relate to the dose order, the administration of the dose order, or a patient which the dose orders to be administered. That is, any data field relevant to the preparation and/or administration of the dose order may be contained in the data stream **105**. The data provided in relation to such data fields may in turn be standardized into a corresponding data field in a staging table that may include predefined data fields to which the dose orders in the data stream **105** are standardized. Additionally or alternatively, a staging table with data fields corresponding to the fields of the dose order in the data stream **105** may be generated upon receipt of the data corresponding to the dose order.

(30) In this later regard, the method may include populating **220** a staging table with the identified/parsed dose order

data from the data stream **105**. For example, the staging table may include data fields stored as a specific data structure corresponding to the standardized intermediate format. For instance, the standardized intermediate format may be defined by a database schema associated with dose orders. In this regard, to the extent dose order metadata for a given dose order data field is parsed with respect to a given one of the dose orders in the data stream **105**, a corresponding staging table instance (e.g., a row in the table and/or a unique staging table entry or record) may be generated and appropriate respective data fields according to the standardized intermediate format may be populated **220** the corresponding dose order metadata identified/parsed **215** from the data stream **105**. In turn, a staging table corresponding to each identified individual dose order may be generated and populated **220** with parsed dose order data. It may be appreciated that the individual staging tables regarding corresponding individual dose orders may be collectively stored in a database. In this regard, individual database files need not be generated for each individual dose order, but rather an identified staging table or portion of a larger staging table (e.g., a unique row or record) may be specifically related to a given dose order for specific identification relative thereto.

(31) The method **200** may, as an option in at least some embodiments, include determining **225** a dose fulfillment client for use in preparation of a given dose order. For example, the dose order data contained in the data stream **105** for a given dose order or plurality of doses may include an express identification of a dose fulfillment client **140** for use in fulfillment of the corresponding dose order(s) (e.g., the healthcare information data system **105** may include an indication of the dose fulfillment client **140** for given a given dose order that originates as data provided by the EMR system **110**). That is, the EMR system **110** may send data indicative of the desired dose fulfillment client **140** to be used to fulfill a particular dose order (e.g., by indicating dose fulfillment client type or by indicating a specific identity of a dose fulfillment client) that is contained in the healthcare data stream **105**. In another example, a source of the data stream **105** may be identified, and the identity of the source of the data stream **105** may at least in part be used to determine the dose fulfillment client **140** to which the dose order is to be provided for fulfillment. For example, all dose orders received from a specific given source may be defined as being dose orders corresponding to a specific dose fulfillment client **140** for use in fulfillment of those specific dose orders from the specific source. Further still, the determination **225** of the dose fulfillment client **140** for a dose order may be based upon one or more portions of dose order data. For example, the dose order data may define a dose characteristic (e.g., an ingredient, ingredient amounts, ingredient types, etc.) that may be used to determine **225** dose fulfillment client **140** for use in fulfillment of the dose order. In this regard, logic may be established in accord with any of the foregoing examples to facilitate determination **225** of the dose fulfillment client **140** for a given dose order. For instance, the logic may be executed by a client router **128** (e.g., shown in FIG. **3** et seq.) that may be provided at the transformation module **130**. In other contemplated embodiments, the client router **128** may be provided remotely from the transformation module **130** (e.g., at the platform interface module **120** and/or as a completely separate module).

(32) Furthermore, the method **200** may, as an option in at least some embodiments, include performing **230** dose management functions with respect to the dose orders received and stored in corresponding staging tables. For example, the dose orders stored in corresponding staging tables may be reviewed, modified, prioritized, grouped, organized, or otherwise managed. It should be noted that the dose management functions that may be performed **230** may occur prior to receipt of dose order data at a dose fulfillment client **140** and/or upon receipt of dose order data at the dose fulfillment client **140**. That is, the dose management functions may be performed by the platform interface module **120**, the transformation module **130**, and/or the dose fulfillment client **140**. For instance, the dose management functions may be performed by a queue management and prioritization module **152**, which may be located at the platform interface module **120** (e.g., as a standalone module and/or as a component of the client router **128** or transformation module **130**), at a dose fulfillment client **140**, and/or disposed between the platform interface module **120** and a dose fulfillment client **140**.

(33) The method **200** may further include transforming **235** dose order data from the standardized intermediate format (e.g., the staging table) into a specific format associated with a dose fulfillment client **140** (e.g., a unique input format for a given dose fulfillment client **140**). As will be described in greater detail below, the transforming **235** may include mapping dose order data fields of the staging table to input fields for a unique input format of a given dose fulfillment client **140**. Furthermore, dose order metadata for a given dose order stored in a dose order data field may be translated into a format associated with the unique input format of the given dose fulfillment client **140**. In this regard, the transformation **235** may be at least in part depending upon the determination **225** of the dose fulfillment client for a given dose order as described above.

(34) The method **200** may further include providing **240** the transformed dose order data to a dose fulfillment client **140**. In this regard, the dose fulfillment client may request the transformed dose order data (e.g., pull the data from the platform interface **120** and/or transformation module **130**) and/or the dose fulfillment client **140** may have the transformed dose order metadata transmitted thereto (e.g., the data may be pushed from the platform interface **120** and/or transformation module **130**).

(35) In any regard, the method **200** may include validating **245** the transformed dose order metadata relative to the unique input format for the dose fulfillment client **140** that receives or is to receive the dose order data. For instance, the validating **245** may include determining whether required dose fulfillment client input fields are present in the

transformed dose order data. Furthermore, the validating **245** may include analyzing a portion of transformed dose order data to determine if a valid corresponding portion of data is available in the unique input format for the dose fulfillment client **140**. In this regard, in the event the validating **245** is unsuccessful, an exception may be generated with respect to a dose order. In turn, the method **200** may include exemption processing to resolve an error associated with the validation **245**. Examples of exception processing are provided below in greater detail.
(36) The method **200** may also, as an option in at least some embodiments, include performing **250** logistical processing with respect to the dose orders. For example, the dose order data stream **105** may include order messages of differing types. For instance, dose order message types may relate to new dose orders, dose order change requests, dose order cancellation requests, or other appropriate dose order requests. That is, at least some of the dose order message types processed may require subsequent actions to be taken on a previously received dose order. Accordingly, in the event dose order data is processed and provided to a dose fulfillment client **140** and a subsequent dose order message is received that requires action be taken with respect to the dose order data provided to the dose fulfillment client **140**, additional logistic processing may be performed **250**. This may include analysis of prior dose orders to determine the state of the previous dose order (e.g. whether the dose order has yet been sent to the dose fulfillment client, whether the dose has been fulfilled, etc.). Additionally, the logistical processing may also include performing actions (e.g., modification, cancellation, etc.) of a prior dose order as required by a subsequently received dose order message. Furthermore, it may be appreciated that a dose order message may apply to a dose order that has been routed to a first dose fulfillment client **140**, but upon the execution of the dose order message is to be recalled from the first dose fulfillment client **140** and provided to a second dose fulfillment client **140**.
(37) The method **200** may also include updating **255** a staging table to reflect receipt of a dose order at the dose fulfillment client **140**. For instance, the updating **235** may include modifying a data field of the staging table associated with an indication of the dose order status to indicate the dose order record has been provided to and/or received by the dose fulfillment client **140**.
(38) The method **200** may further include fulfilling **260** a dose at the dose fulfillment client **140**. The fulfilling **260** may include preparation of a dose corresponding to a dose order at the dose fulfillment client **140**. The preparation may be a manual operation to prepare the dose order, an automated operation to fulfill the dose order, or a combination thereof. Further still, the fulfilling **260** may include making available a dose (e.g., a pre-prepared) to satisfy a dose order (e.g., such as providing instructions to a dose dispensation cabinet to make a prepared dose available for fulfillment of the dose order).
(39) With further reference to FIG. **3**, a schematic depiction of an embodiment of a system **300** for automated healthcare exchange is depicted. The system **30** may include an EMR system **110** that provides a healthcare information data stream **105** in the form of an HL7 data feed to a HL7 data feed port **122** of the platform interface module **120**. In turn, the HL7 data received at the HL7 data feed port **122** may be provided to an HL7 parsing module **124**. In this regard, the HL7 parsing module **124** may identify/parse dose orders from the HL7 data feed received at the HL7 data feed port **122**. As such, the HL7 parsing module **127** may comprise a stream processing module. The HL7 parsing module **124** may also determine if the dose order identified form the data stream **105** is to be processed by the exchange system **30**. For instance, a dose may be processed by the system as described herein or may be diverted to a different dose handling system by the parsing module **124** based on the dose order identified.
(40) In any regard and as described above, a staging table may be populated for a dose order identified/parsed from the data stream **105**. The staging table may be a predefined data structure that is populated with identified/parsed data from the data stream **105**. In turn, the dose order data may be used to populate fields in the staging table corresponding to dose order data fields with dose order metadata from the data stream **105**. In turn, the staging table regarding the dose order may be stored in a staging table database **126**.
(41) The staging table database **126** may be in operative communication with a central server **300** that is disposed remotely from the platform interface module **120**. In this regard, data regarding the dose orders stored as staging tables in the staging table database **126** may be provided to the central server **300**. The central server **300** may receive dose order data from a plurality of platform interface modules **120** (e.g., from different healthcare facilities implementing platform interface module **120**). The central server **300** may facilitate backup and/or data aggregation services in relation to the dose order data.
(42) The staging table database **126** may be in operative communication with a transformation module **130**. In this regard, the transformation module **130** may be operative to retrieve a staging table from the staging table database for transformation of the dose order data contained therein. Additionally, the transformation module **130** may include a client router **128**. In turn, the client router **128** be operative to provide transformed dose order data (e.g., in a unique input format for a given dose fulfillment client **140**) to an appropriate dose fulfillment client **140**. While shown as a common module, the client router **128** may also be provided separately from the transformation module **130** (e.g., as a standalone module or as a module provided in connection with the platform interface module **120**).
(43) As described briefly above, various methods for determining an appropriate dose fulfillment client **140** to which the dose order is to be directed may be applied by the client router **128**. In this regard, the client router **128** may employ any one or more of the approaches described above to determine an appropriate dose fulfillment client **140** to

which the transformed dose order data may be provided. In this regard, shown in FIG. **3**, a plurality of clients **140**A, **140**B, and **140**C may be provided in operative communication with the client router **128**. In turn, the client router **128** may provide an appropriate one of the fulfillment clients **140**A, **140**B, **140**C with the transformed dose order data such that the appropriate dose fulfillment client **140** may be used to fulfill the dose corresponding to the dose order.
(44) As may be appreciated, the determination of the appropriate dose fulfillment client **140** to which the dose order will be provided may affect the transformation of the dose order by the transformation module **130**. In this regard, the transformation module **130** may comprise the client router **128** as depicted or may be in bidirectional communication with the client router **128** such that the transformation module **130** and client router **128** may collectively act upon the data in a staging table to determine the dose fulfillment client **140** to which the dose order will be prepared such that the identity of the determined dose fulfillment client **140** may at least in part affect the transformation applied to the dose order data by the transformation module **130**. For instance, upon identification of an appropriate dose fulfillment client **140** by the client router **128**, a unique input format for the identified dose fulfillment client **140** may used to transform the dose order data from the staging table.
(45) In other embodiments (e.g., such as that depicted in FIG. **5** discussed in greater detail below), the transformation module **130** of the embodiment of the system **30** may be provided at the platform interface module **120**. Furthermore and as addressed in greater detail below in relation to other embodiments, the client router **128** may be located the platform interface module **120**. In this regard, the fulfillment clients **140**A, **140**B, and **140**C may each be operative to receive transformed dose order data for use in fulfillment of the dose corresponding to the dose order. That is, in at least some embodiments, all processing regarding the dose order data for exchange with the dose fulfillment client **140** may occur at the platform interface module **120**. However, in other embodiments, such as those described below, different functionality with respect to the transformation module **130** and/or client router **128** may be provided in a distributed manner remotely from the platform interface module **120**.
(46) The platform interface module **120** may also include a web server **310** that provides for remote access to the functionality provided by the platform interface module **120**. For example, the web server **310** may maintain and/or execute one or more web pages **312** corresponding to the various modules and/or functionality provided by the platform interface module **120**. In turn, the web pages **312** may facilitate functionality in connection with the platform interface module **120** and/or modules provided comprising the platform interface module. For instance, various settings, parameters, or other administrative functions may be performed with respect to platform interface module **120**. Such functionality may be facilitated by way of the web pages **312** that are accessed by way the web server **310**.
(47) With further reference to FIG. **4**, another embodiment of a system **40** for automated exchange of healthcare information is depicted. The system **40** may be particularly useful in the context of fulfillment of total parenteral nutrition (TPN) dose orders received from the EMR system **110**. The system of FIG. **4** may be configured to receive TPN dose orders from the EMR system **110** in a plurality formats. For example, the platform interface module **120** may include an HL7 data feed port **122** to which HL7 format messages from the EMR system **110** may be routed and parsed as described above in connection with FIG. **3**. The platform interface module **120** may also include a TPN print data feed **402** to which print feed data messages from the EMR system **110** may be routed. In this regard, the platform interface module **120** may facilitate both the capability to process HL7 format messages as well as print feed messages received from the EMR system **110**. Examples of HL7 message formats that may be processed may include, but are not limited to, HL7 messages originating from EMR systems provided by Cerner Corporation, Epic Systems Corporation, Meditech, Omnicell, Inc., or others. Furthermore, examples of supported print feed formats may include, but are not limited to, Zebra programming language provide by Zebra Technologies, DataMax format provided by DataMax-O'Neil, Intermec format provided by Intermec, Inc., portable document format (PDF) provided by Adobe Systems, plain text format, etc.
(48) In this regard, HL7 messages may be provided from the HL7 data feed port **122** to an HL7 parsing module **124** as described above in relation to FIG. **3**. Additionally, the TPN print data feed **402** may provide print feed messages to a label processor **404** of the platform interface module **120**. In any regard, the HL7 parsing module **124** and/or the label processor **404** may provide individual dose order metadata in relation to corresponding dose order data fields for storage as a staging table in a staging table database **126**.
(49) In the example provided in FIG. **4**, the platform interface module **126** may be in direct communication with a TPN fulfillment client **142**. In this regard, the TPN fulfillment client **142** may be a dose fulfillment client **140** capable of fulfilling TPN dose orders. For instance, the TPN fulfillment client **142** may include a TPN management module **144**. The TPN management module **144** may facilitate dose order calculation and management. For instance, the TPN management module **144** may comprise ABACUS Calculation Software provided by Baxter Healthcare Corporation of Deerfield, IL. In any regard, the TPN fulfillment client **142** may provide further processing in relation TPN dose orders in connection with the fulfillment of TPN dose orders (e.g., calculations related to ingredient interactions, etc.). The TPN management module **144** may be in further communication with a TPN compounder **158**. The TPN compounder **158** may be operative to autonomously prepare a dose corresponding to a TPN order. In this regard, in an embodiment, the dose fulfillment client **140** may comprise the TPN fulfillment client **142** that collectively

includes the TPN management module **144** and/or the TPN compounder **158**. That is, the transformation module **130** may communicate directly to the TPN management module **144** or the TPN compounder **158**. As such, the unique input format utilized by the transformation module **130** may correspond to the TPN management module **144** or the TPN compounder **158**.

(50) In this regard, and as depicted in FIG. **4**, a transformation module **130** may be provided in operative communication with the staging table database **126** and the TPN client **142**. In this regard, the transformation module **130** may access the staging table database **126** to retrieve dose order data in the standardized intermediate format of the staging table that is transformed by the transformation module **130** into a unique input format for use with the TPN client **142**. In this regard, the system **40** of FIG. **4** may not include a client router **128**. For instance, the platform interface module **120** may be utilized in a dedicated manner to receive only TPN dose orders from the EMR system **110**. In this regard, the TPN client **142** may access the staging table database **126** to retrieve all staging tables contained therein for transformation by the transformation module **130** into the unique input format associated with the TPN management module **144**. That is, the platform interface module **120** may be used in a dose context-specific application wherein only a single type of dose orders to be fulfilled by a given dose fulfillment client **140** (e.g., the TPN fulfillment client **142**) are processed by the platform interface module **120**.

(51) FIG. **5** depicts another embodiment of a system **50** for exchange of healthcare information. The system **50** may also receive HL7 format messages as well as print feed messages from an EMR system **110** that are processed by an HL7 data feed port **122** and a print feed port **402**, respectively, as described above in relation to FIG. **4**. As described above, data messages may be routed to an appropriate one of an HL7 parsing module **124** or a label processor **404** for generation of a staging table stored in the staging table database **126** for corresponding receive dose orders. FIG. **5** further includes a manual order entry website **312**A (e.g., as provided in websites **312** access by web server **310** of FIG. **3**). In this regard, user may access the platform interface module **120** directly using the manual order entry website **312**A to generate a dose order directly at the platform interface module **120** that are stored in a staging table in the staging table database **126**.

(52) In the embodiment of the system **50** depicted in FIG. **5**, platform interface module **120** may be in operative communication with a pharmacy workflow management application **146**. In this regard, the pharmacy workflow management application **146** may be the only dose fulfillment client **140** with which the platform interface module **120** is in operative communication. In this regard, as depicted in FIG. **5**, the transformation module **130** may be provided with platform interface module **120** for transformation of dose order data stored in staging tables of the staging table database **126** into a unique input format associated with the pharmacy workflow management application **146**. Upon receipt of the transformed dose order data, the data may be stored in the dose order record database **148** of the pharmacy workflow management application **146**. In turn, the dose orders may be provided to a pharmacy workflow management module **150**. The pharmacy workflow management module **150** may allow for manual preparation of a dose order by a pharmacy technician or the like. For example, embodiments of a pharmacy workflow management application **146** is described in the commonly assigned U.S. provisional Application No. 62/057,906 filed on Sep. 30, 2014 entitled "MANAGEMENT OF MEDICATION PREPARATION WITH FORMULARY MANAGEMENT', the entirety of which is incorporated by reference herein, may be utilized to fulfill dose orders at the pharmacy workflow management application **146**.

(53) With further reference to FIG. **6** another embodiment of a system **60** for automated healthcare information exchange is depicted. The system **60** may receive dose order information from an EMR system **110** by way of an HL7 data feed port **122**, a print data feed port **402**, or by way of manual order entry from a manual order entry website **312**A as described above in connection to FIG. **5**. In this regard, dose order data may be received and used to populate corresponding staging tables that are stored in the staging table database **126**.

(54) The system **60** of FIG. **6** further includes a plurality of dose fulfillment clients **140**. For instance, the platform interface module **120** is in operative communication with a pharmacy workflow management application **146** and a TPN fulfillment client **142**. The system **60** of FIG. **6** may have a local transformation module **130**A provided at the platform interface module **120**. The local transformation module **130**A may perform data transformation on dose order data that are provided to the pharmacy workflow management application **146**. Additionally, a queue management and prioritization module **152** may be provided for management of dose orders corresponding to dose order data that are to be provided or that are provided to the pharmacy workflow management application **146**.

(55) Additionally, the TPN fulfillment client **142** may have associated therewith a remote transformation module **130**B located remotely from the platform interface module **120** for transformation of dose order data locally with respect to the TPN fulfillment client **142** for dose order data corresponding to dose orders directed to the TPN fulfillment client **142**. In this regard, the system **60** of FIG. **6** may have different transformation modules **130**A and **130**B for transformation of dose order data provided to the pharmacy workflow management application **146** and the TPN fulfillment client **142**, respectively. Furthermore, a queue management and prioritization module **152** may be provided for at least a portion of the dose orders being directed to a given one of the dose fulfillment clients (e.g., the pharmacy workflow management application **146**). The queue management and prioritization module **152** may be operative to manage dose orders prior to or after data transformation has occurred with respect thereto. Thus, while

shown as residing between the transformation module **130** and the pharmacy workflow management application **146** in FIG. **6**, one or more queue management and prioritization modules **152** may be provided elsewhere for management of dose order queues (e.g., at the staging table database **126**, at the dose fulfillment client **140**, etc.).

(56) As described above briefly, the determination of the fulfillment client **140** to which the dose order data may be provided may occur in a number of manners. For example, each respective client may access the staging table database **126** and request dose order data that are flagged (e.g., include a data field indicative of) as being designated for the given dose fulfillment client **140** requesting the information. Furthermore, logic may be applied to the dose order data based on a number of different potential parameters to identify a dose fulfillment client **140** for use in fulfillment of a given dose order. In this regard, the fulfillment clients **140** may be provided dose order data for use in fulfillment of the dose order based on a data field in which the designated dose fulfillment client **140** is indicated as determined by application of the logic.

(57) With further reference to FIG. **7**, an embodiment of a system **70** for automated exchange of healthcare information is depicted. The system **70** may receive dose order data from an EMR system **110** as described above. The system **70** may also include a transformation module **130**D that facilitates transformed data being provided to one or more of a plurality of dose fulfillment clients **140** (e.g., comprising the TPN client **142**, a dispending cabinet **154**, or an automated syringe filler **156** in the depicted embodiment). The transformation module **130**D may include a client router **128**.

(58) In this regard, the dose fulfillment clients **140** of the system **70** may include a pharmacy workflow management application **146**, a TPN client **142**, a dispensing cabinet **154** (e.g., an automated dose dispending cabinet), and/or an automated syringe filler **156**. A portion of the dose orders from the staging table database **126** may be provided to the pharmacy workflow management application **146** by way of transformation module **130**. Another portion of dose orders may be provided to the transformation module **130**D. In turn, the client router **128** of the transformation module **130**D may be operative to identify a dose fulfillment client **140** to which the dose order should be provided. As such, the client router **128** may apply any appropriate logic with respect to the dose order to determine the appropriate dose fulfillment client **140**. For example, the client router **128** may identify and utilize a dose fulfillment client flag in the dose order has received from the EMR system to direct the dose order to an appropriate dose fulfillment client **140**. Furthermore, the client router **128** may dynamically route the dose order to an appropriate fulfillment client **140** based on the dose order metadata contained by the dose order. Specifically, the client router **128** may be operative to determine whether the dose order is appropriate for automated preparation (e.g., by an automated syringe filler **156**). In this regard, if the dose order is appropriate for automated preparation, the client router **128** may provide the dose order to an appropriate automated dose fulfillment client **140** (e.g., such as automated syringe filler **156**). Further still, the client router **128** may be operative to identify if the dose order is appropriate for fulfillment by way of an automated dispensing cabinet **154**.

(59) As may be appreciated, it may be depended upon the specific dose fulfillment client **140** to which a dose order is provided as to the processing of the specific dose order data. For example, upon provision of dose orders to one of the pharmacy workflow management application **146** transformation module **130**C that is provided locally to the platform module interface **120** may allow for transformation of the dose order data in the staging table database **126** into a unique input format corresponding to pharmacy workflow management application **146**. Furthermore, the transformation module **130**D may be provided (e.g., remotely from the platform interface module **120**) for transformation of dose order data with respect to the TPN fulfillment client **142**, dispending cabinet **154**, or automated syringe filler **156**. In this regard, the TPN fulfillment client **142** may include a TPN compounder **158** for compounding of the TPN dose order according to the dose order data received at the TPN client **142** and transformed by the transformation module **130**D. Furthermore, the TPN order management module **144** may provide information to the pharmacy workflow management application **146**. While shown as a direct connection, it is anticipated that the dose order data may be provided by way of the platform interface module **120** and/or transformation module **130**C). Further still, the dose order data may be provided to both the pharmacy workflow management application **146** and the TPN fulfillment client **142** and an indication of a correlation between the dose order data provided to each respective dose fulfillment client **140**, respectively.

(60) FIG. **8** further depicts a system **80** that, like the system **70** in FIG. **7** is in operative communication with a pharmacy workflow management application **146**, a TPN client **142**, a dispensing cabinet **154**, and/or an automated syringe filler **156**. However, unlike the system **70** in FIG. **7**, the system **80** in FIG. **8** may include a single transformation module **130** provided at the platform interface module **120** for transformation of dose order data from the standardized intermediate format of the staging table stored in the staging table database **126** into a corresponding unique input format for a given dose fulfillment client **140**. In this regard, the platform interface module **120** may include a client router **128** that is disposed between the staging table database **126** and the EMR system **110** or other means of order entry. In this regard, the client router **128** may be operative to apply logic to determine an appropriate one of the dose fulfillment clients **140** for use with each respective one of the dose orders. In this regard, the client router **128** may append information to the data in the staging table for a given dose order indicative of a dose fulfillment client **140** to which the dose order is to be provided. Further still, the client router **128** may be in operative

communication with transformation module **130** to provide information regarding the dose fulfillment client **140** to which a dose order for a given staging table may be provided.

(61) In any of the foregoing embodiments, the respective systems for automated transformation healthcare information may also facilitate logging with respect to actions taken relative to a dose order. The logging may include generating logs that reflect the actions taken on the dose order. The logs may be maintained for later review by human user (e.g., in the performance of an audit, troubleshooting, or the like). The logging may occur, and the logs may be stored, at different respective portions within the system. For example, the platform interface module **120** may perform logging to generate corresponding logs. Furthermore, the transformation module **130** (e.g., regardless of their specific location within the systems) may also perform logging to generate corresponding logs. In this regard, the logs may be maintained at different respective portions within the system. Additionally or alternatively, the logs may be transmitted within the system to a common location. For example, the platform interface module **120** may be in operative communication with transformation modules **130**, or dose fulfillment clients **140** to receive logs there from regarding processing in relation to dose orders. In this regard, the logs for one or more portions of the system may be collectively stored at a given point in the system such as, for example, the platform interface module **120**.

(62) With further reference to FIG. **9**, a representation of a data transformation performed by a transformation module **130** is shown. Specifically, FIG. **9** includes a textual representation of the data transformation in human perceivable form. FIG. **9** includes a representation of an embodiment of a date feed **500**. As may be appreciated, the data feed **500** may be analyzed (e.g., by a parsing module of the platform interface module **120**). The parsing may result in a dose order **502** and a dose order **504** being identified. As may be appreciated, a portion **518** of the data stream **500** may not correspond to a dose order.

(63) Dose order **502** may be parsed such that a number of dose order data fields are recognized. Specifically, the dose order **502** may include a first patient data field **506**, a second patient data field **508**, a first ingredient field **510**, a second ingredient field **512**, an administration data field **514**, and a patient procedure data field **516**. Additional or fewer data fields may be present such that those presented are for demonstrative purposes only. The second dose order **504** may be parsed such that a first data patient field **524**, a first ingredient field **526**, and an administration data field **528** are identified. In this regard, the data for the first dose order **502** and the second dose order **504** may be stored in respective staging tables. As such, the textual representation of the first dose order **502** may correspond to the staging table for the first dose order **502** and the textual representation of the second dose order **504** may correspond to the staging table for the second dose order **504**. While depicted textually in FIG. **9**, it may be appreciated that the actual staging tables may be maintained in a different format such as an XML format, a SQL database format, or other appropriate format.

(64) In any regard, the transformation module **130** may be operative to map data fields from the staging tables **502** and **504** to corresponding fields in unique input formats for one or more dose fulfillment clients **140**. For instance, a first input **552** may be generated that corresponds to the staging table for the first order **502**. As may be appreciated, the first input **552** may have defined fields according to a unique input format for a specific dose fulfillment client **140**. As such, appropriate ones of the data fields in the staging table **502** may be mapped to fields in the first input **552**. Specifically, the first patient data field **506** and the second patient data field **508** may be mapped to the client patient info field **556** of the first input **552**. In this regard, it may be appreciated the multiple data fields from the staging table **502** may be mapped to a single field in the first input **552**. In this regard, the data from the multiple fields (e.g., the first patient data field **506** and the second patient data field **508**) of the staging table **502** may be aggregated or reformatted for inclusion in the client patient input **556**. Furthermore, a portion of the data contained in one of the first patient data field **506** or second patient data field **508** may be omitted.

(65) Further examples of the matching between data fields in the staging tables **502**, **504** and the first and second input **552**, **554** provided in FIG. **9**. Specifically, the first ingredient field **510** and the second ingredient field **512** may be mapped to a dose ingredient input **558** of the first input **552**. Furthermore, the administration data field **514** of the first staging table **502** may be mapped to the administration data input **560** of the first input **552**. Of note, not all data field of the first staging table **502** may be mapped to the first input **522**. For example patient procedure data field **516** may be irrelevant for the dose order, and therefore not have a corresponding input field of the first input **552** to which the patient procedure data field **516** is mapped. Furthermore, while the first staging table **502** demonstrates that multiple data fields from the staging table **502** may be mapped to a single info field in the first input **552**, it may also be appreciated that one to one correspondence may be provided such as in the case of the second dose order **504**. In this regard, the first patient data field **524** of the second dose order **504** may be mapped to the client patient input **562** of the second input **554**. Furthermore, the first ingredient field **526** may be mapped to the dose ingredient input **564**, and the administration data field **528** may be mapped to the administration data input **566**. Furthermore, it should be noted that portion **518** of the data stream not corresponding to a dose order may not be mapped to any client input. That is, at least one platform interface module **120** or transformation module **130** may recognize the portion **518** is not corresponding to the dose order such that no corresponding input is generated or mapped to the portion **518**.

(66) In addition to the mapping of data fields corresponding to dose orders to client input fields, the transformation module **130** may also perform translation with respect to the data contained in each respective field such that the data

within each respective field may be translated into a format or form expected by the client in a given client info field. For instance, with further reference to FIG. **17**, a user interface **1700** that includes representations of potential translations to be performed by the transformation module **130** is provided. The user interface **1700** may include a type listing **1710**, a client form listing **1720**, and an EMR form listing **1730** provided in a table **1702**. In this regard, the table **1702** may include a plurality of listings (e.g., arranged by type **1710**) for translation from a EMR form **1730** to a client form **1720**. As may be appreciated, some listings may provide identical forms for the EMR form **1730** and the client form **1720**. For example, the form corresponding to listing **1712** for "Acetate" may be the same for the EMR form **1732** and the client form **1722**. However, as may be appreciated from the listing **1702**, other forms may differ between the EMR form **1730** and the client form **1720**. For example, listing **1714** may have a EMR form of "ADULTTRACE" and a client form **1724** of "Adult Trace Conc." Accordingly, should the EMR form **1734** for listing **1714** appear in a data field for a staging table, the data may be translated to the client form **1724** by the transformation module **130**. In this regard, the user interface **1700** may also include a type filter **1740** and a client form filter **1750** that may be used to narrow the results presented in the table **1702**. Furthermore, upon selection of a given listing within the table **1702**, the EMR form **1730** for a given listing may be modified using the EMR form input **1760**. For instance, upon selection of a listing from the table **1702** for a given portion of data, the EMR form **1730** may be modified for the listing using the EMR form input **1760**. Furthermore, listings may be added or removed from the listing **1702** by the user. In this regard, it may be appreciated that the user may be able to maintain the definitions provided in the listing **1702** regarding translations to be applied by the transformation module **103** using the user interface **1700**. In this regard, the user interface **1700** may be provided by a webpage **312** of the platform interface module **120** and/or may be provided by a dose fulfillment client **140** that comprises a transformation module **130** as described above.

(67) With further reference to FIGS. **10-12**, an example of the processing of a dose order from an EMR system **110** to a dose fulfillment client **140** is depicted. Specifically, FIG. **10** depicts a textual representation of the data stream **600** from which the dose order is identified. In turn, FIG. **11** presents a textual representation of a log maintained by a transformation module **130** when acting upon the identified dose order from the data stream **600**. FIG. **12** represents a user interface **800** of a dose fulfillment client **140** that is received the dose order. In this regard, the textual representation of the data stream **600** may include a plurality of data fields. For example, highlighted data fields corresponding to a patient data field **610**, a dose order type field **620**, a macro ingredients detail field **630**, and a micro ingredients detail field **640** that are identified for a given dose order from the data stream **600**. In this regard, the metadata for each corresponding field may be stored in a staging table. For example, the metadata found within the patient data field **610** may be used to populate one or more fields in a staging table related to the patient. Specifically, the name "TONI SMC," a gender "female," a date of birth "19871229," and an EMR number "123456" may be identified in the patient data field **610**. Furthermore, an identifier indicating the dose order is a TPN order may be identified from the dose order type field **620**. Furthermore, a macro ingredient **630** "AMINOSYN II 15% IV SOLN{circumflex over ( )}RX|100.8|g|100.8|g" for the dose order **630** may be contained within the macro ingredient data field **630**. Furthermore, to micro ingredients "SODIUM CHLORIDE 4 MEQ/ML IV SOLN{circumflex over ( )}RX|33.6|MEQ|33.6|MEQ" and "POTASSIUM ACETATE 2 MEQ/ML IV SOLN{circumflex over ( )}RX|16.8|MEQ|16.8|MEQ" may be identified in the micro ingredients detail field **640**.

(68) With further reference to the log **700** generated by the transformation module **130**, it may be appreciated that the transformation module **130** may convert the patient data field **610** into an input for creation of a patient in the dose fulfillment client **140** (e.g., in this case an ABACUS Calculation Software application) at line **710**. Furthermore, the dose order type field **620** may be used to create an input for the creation of an order at the dose fulfillment client **140** at line **720**. In turn, the order for the dose fulfillment client **140** may be populated with inputs corresponding to the macro ingredient detail **630** and the micro ingredient detail **640** such that those data fields are represented in the order input data **734** the dose fulfillment client **140**. As may be appreciated, the macro ingredient detail field in the micro ingredient detail field **630** and **640** may be transformed into a format associated with unique input format associated with the dose fulfillment client **140**. In this regard, the input fields **630′** and **640′** corresponding to the macro ingredient detail field **630** and the micro ingredient detail field **640** may contain data in a translated format. In turn, the inputs to the dose fulfillment client **140** reflected in the log **700** may be used to provide inputs to the dose fulfillment client **140**.

(69) In turn, FIG. **12** displays a user interface **800** of the dose fulfillment client **140** after receiving the commands reflected in the log **700**. As may be appreciated, patient information file **10** may be presented with information corresponding to the data contained in the patient data field **610**. Specifically, the patient name, the EMR number, an age based upon the date of birth provided in the patient data field **610**, or other information may be provided in the patient information field **810**. Furthermore, in ingredient listing **820** of the dose fulfillment client **140** may be provided. As may be appreciated, the macro ingredient **830** listed in the ingredient listing **820** corresponds to the data contained in the macro ingredient data field **630**. The micro ingredients **840** listed in the ingredient listing **820** correspond to the data contained in the micro ingredient data field **840**. As may be appreciated, the form of the ingredients **830** and **840** may be different from that contained in the data fields **630** and **640**, thus reflecting the

translation performed by the transformation module **130**.

(70) With further reference to FIG. **13**, the user interface **900** for a dose fulfillment client **140** is depicted. The user interface **900** may include a patient listing **910**. The patient listing **910** may list patients for which dose orders are pending. Upon receipt of a dose order from a transformation module **130**, the dose fulfillment client **140** may update the patient listing **910** with the status indicator **912** to indicate that the dose order has been received. In this regard, the user interface **900** may further include an order listing field **920**. The order listing field **920** may list complete and pending dose orders for a given patient selected from the patient listing **910**. As may be appreciated, a first pending order **922** and second pending order **924** may be indicated as having been received by the dose fulfillment client **140**. Status indicators for the received pending doses **922** and **924** may be provided. Specifically, a first status indicator **926** may be provided to indicate that no exceptions were generated in connection with the processing of the first pending order **922** by the platform interface module **120** and/or transformation module **130**. In this regard, the first status indicator **926** may indicate the first pending dose order **922** is ready to be fulfilled by the dose fulfillment client **140**.

(71) The second status indicator **928** provided with the second pending dose order **924** may indicate that an exception was generated during the processing of the second pending dose order **924**. Upon selection of the second pending dose **924**, a dose detail screen **950** depicted in FIG. **14** may provided to the user. The dose detail screen **950** may include a patient information field **810** is described above in connection with FIG. **12**. Furthermore, the dose detail screen **950** may include an ingredient listing **820** as described above in connection with FIG. **12**. The dose detail screen **950** may also include an exception field **960** that may list exceptions identified during the processing of the dose order by the platform interface module **120** and/or transformation module **130**. For example, exceptions may be identified with respect to an unrecognized product, an unrecognized value, and unrecognized unit, and unrecognized data field, or other error associated and the processing of the dose order. For instance in FIG. **14**, an exception **962** may be provided corresponding to an error that occurred with respect to the mapping of an ingredient “Insulin” to the dose fulfillment client **140**. In this regard, the exception listing **960** may include an EMR form, a substance description, a value, and a unit associated with the exception **962**. Accordingly, the user may select the exception **962** and resolve the error associated therewith. For example, in FIG. **15**, the dose detail screen **950** is shown such that the user has added a further ingredient **964** to the ingredient listing **820** associated with the ingredient that is the subject of the exception **962**. In this regard, the user may flag the exception **962** as having been processed and made proceed with preparation of the dose by the dose fulfillment client **140**.

(72) Furthermore, in FIG. **16** a user interface **1000** may be provided for use in resolution of exceptions, such as the exception **962** identified in FIGS. **14** and **15**. For example, the exception **962** may be caused by an improper mapping between the EMR form for the drug “Insulin” and the client input form associated with the dose fulfillment client **140**. That is, the mapping between the EMR form and the client input form may have caused an error when the platform interface module **120** and/or the transformation module **130**A to process the data field containing the data related to insulin for this dose order. In this regard, the user interface **1000** may allow user to select the product **1002** for modification by the user interface **1000**. In turn, the user may be operative to modify the formulary record for the selected product **1002**.

(73) Specifically, the user interface **1000** may also include a EMR correspondence field **1004** that may be used to provide the corresponding EMR form for the product **1002** at the dose fulfillment client **140**. In this regard, the user may use the user interface **1000** to rectify the error in mapping between the EMR form and the client form for the drug “insulin”. In turn, upon further processing of the subsequent dose order containing the ingredient “insulin”, the appropriate translation may be provided such that the exception **962** may not occur in a subsequent order having the same product **1002** provided therewith. While the user interface **1000** provides for exception resolution at the dose fulfillment client **140**, the exception may also be flagged at the platform interface module **120** and/or the transformation module **130** in appropriate user interfaces that may be provided to facilitate provision of a correct mapping and/or translation for a product in a dose order in a manner similar is that discussed with respect to FIG. **16**. That is, the exception processing described in relation to FIG. **16** from the perspective of the dose fulfillment client **140** may also be provided in connection with the platform interface module **120** and/or transformation module **130**. Furthermore, while ingredient specific processing may be fulfilled using the user interface **1000**, bulk changes to the mapping between EMR forms **1730** and client forms **1720** may be facilitated by the user interface **1700** described above in connection with FIG. **17**.

(74) While the invention has been illustrated and described in detail in the drawings and foregoing description, such illustration and description is to be considered as exemplary and not restrictive in character. For example, certain embodiments described hereinabove may be combinable with other described embodiments and/or arranged in other ways (e.g., process elements may be performed in other sequences). Accordingly, it should be understood that only the preferred embodiment and variants thereof have been shown and described and that all changes and modifications that come within the spirit of the invention are desired to be protected.

Claims

1. A method for an automated exchange of healthcare information between an electronic medical record (EMR) system and a selected dose fulfillment client for fulfillment of a dose order, the method comprising: executing non-transitory machine-readable data for specifically configuring processors of a platform interface module, a transformation module, and a client router, execution of the non-transitory machine-readable data causing the processors to: receive a healthcare information data stream at the platform interface module via a network from the EMR system, wherein the healthcare information data stream comprises information including dose order metadata related to dose orders that is provided in a Health Level 7 (“HL7”) format; identify, via the platform interface module, individual dose orders from the healthcare information data stream by identifying express delineators within the healthcare information data stream and identifying the dose order metadata corresponding to dose order data fields, wherein the express delineators mark separations between the dose orders in the healthcare information data stream; parse, via the platform interface module, the dose order metadata for each of the dose orders from the healthcare information data stream, wherein the dose order metadata comprises one or more dose characteristics associated with the respective dose order; populate, via the platform interface module, a staging table with the dose order metadata for each of the dose orders, wherein the staging table comprises a plurality of dose order data fields populated with corresponding respective portions of the dose order metadata parsed from the healthcare information data stream, and wherein the dose order metadata is populated in a standardized intermediate format; determine, via the transformation module from the dose order metadata, a dose fulfillment client type, among a plurality of different dose fulfillment client types for fulfilling each of the dose orders, the dose fulfillment client types including an automated dose preparation device, an automated total parenteral nutrition (TPN) compounder, and an automated dose dispensing cabinet; select, via the transformation module, a dose fulfillment client that matches the dose fulfillment client type for fulfillment of each of the dose orders; transform, via the transformation module, the dose order metadata from the standardized intermediate format into a predefined format dictated by the selected dose fulfillment client to generate transformed dose order metadata, wherein the transforming is at least in part based on the respective predefined format of the selected dose fulfillment client, and wherein the transforming includes mapping the plurality of dose order data fields of the staging table to corresponding respective ones of a plurality of dose fulfillment client input fields for the selected dose fulfillment client, including at least: mapping one data field of the plurality of dose order data fields of the staging table to one data field of the plurality of dose fulfillment client input fields, and mapping at least two data fields of the plurality of dose order data fields of the staging table to one data field of the plurality of dose fulfillment client input fields; route, via the client router of the transformation module, the transformed dose order metadata to the selected dose fulfillment client via the network; and cause each of the dose orders to be fulfilled automatically using the selected dose fulfillment client based on the transformed dose order metadata to provide a corresponding medication dose.
2. The method of claim 1, further comprising: receiving a first dose order in a first EMR form; and receiving a second dose order in a second EMR from, wherein the first dose order and the second dose order correspond to a dose order with identical constituent ingredients, wherein a form of the constituent ingredients in the first EMR form differs from the second EMR form, and wherein a form of the constituent ingredients for the first dose order is identical to a form of the constituent ingredients for the second dose order in the respective staging tables corresponding to the first dose order and the second dose order.
3. The method of claim 1, wherein the staging table is independent of any of the plurality of dose fulfillment clients.
4. The method of claim 1, wherein the dose fulfillment client input fields are defined by a unique input format associated with the dose fulfillment client.
5. The method of claim 4, wherein the transforming comprises: validating the dose order metadata of the plurality of dose order data fields with respect to the unique input format for corresponding ones of the dose fulfillment client input fields.
6. The method of claim 5, wherein the validating comprises: correlating the dose order metadata of the plurality of dose order data fields to formulary records of the dose fulfillment client with respect to the corresponding respective ones of the dose fulfillment client input fields.
7. The method of claim 6, further comprising: generating an exception in response to an error associated with at least one of the mapping or validating.
8. The method of claim 7, wherein the exception comprises prompting a human user to resolve the error.
9. The method of claim 8, further comprising: receiving, from the human user, an input associated with the resolution of the error, wherein the input comprises at least one of a correct mapping between a dose order data field of the staging table and a corresponding respective one of a dose fulfillment client input fields or a correct correlation between a portion of dose order metadata and a formulary record of the dose fulfillment client.
10. The method of claim 9, further comprising: updating at least one of a mapping logic or a correlation logic for use in the mapping and correlating, respectively, based on the input received from the human user.

11. The method of claim 1, further comprising: updating the staging table to indicate the dose order metadata for each of the dose orders that have been routed to the dose fulfillment client.
12. The method of claim 1, further comprising: managing each of the dose orders prior to providing the dose order metadata in a second form to the dose fulfillment client.
13. The method of claim 12, wherein the managing includes providing a user interface to allow a user to perform a management function relative to each of the dose orders.
14. The method of claim 13, wherein the management function comprises at least one of modification of dose order metadata, cancellation of the dose order, organization of the dose order relative to other dose orders, or prioritization of the dose order relative to other dose orders.
15. The method of claim 1, further comprising: performing logistical processing on each of the dose orders.
16. The method of claim 15, wherein the logistical processing comprises performing an action relative to each of the dose orders in response to receipt of an EMR system message subsequent to the receipt of the respective dose order.
17. The method of claim 16, wherein the EMR system message comprises at least one of a dose order change message or a dose order discontinuation message.
18. The method of claim 17, wherein the logistical processing comprises determining if the dose order metadata has been provided to the dose fulfillment client.
19. The method of claim 18, wherein the logistical processing comprises performing an action relative to each of the dose orders corresponding to the EMR system message for a dose order that has not yet been provided to the dose fulfillment client.
20. The method of claim 19, wherein the logistical processing comprises providing data to the dose fulfillment client related to the EMR system message for a dose order that has been provided to the dose fulfillment client.
21. The method of claim 20, wherein the data provided to the dose fulfillment client related to the EMR system message is transformed into a form corresponding to the dose fulfillment client.
22. The method of claim 1, further comprising: logging activity taken with respect to each of the dose orders to generate a log regarding activities relative to the respective dose order.
23. The method of claim 22, wherein the logging comprises aggregating a plurality of logs generated relative to different respective actions taken with respect to the respective dose order.
24. The method of claim 1, wherein the healthcare information data stream is received from the EMR system.
25. The method of claim 24, wherein the EMR system comprises a hospital information system (HIS).